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METHODS AND COMPOSITIONS FOR PLANT PATHOGEN RESISTANCE IN PLANTS

Abstract

The disclosure relates to a plant that is tolerant or resistant to species of *Ca. Liberibacter.* Specifically exemplified are *Citrus* and solanaceous plants. Provided by the disclosure is a modified *Citrus* or solanaceous plant that is resistant or tolerant to Sec-dependent effectors secreted by bacteria. Also provided by the disclosure are methods of modifying a plant genome plant to provide tolerance or resistance to species of *Ca. Liberibacter.* Still further provided by the disclosure are methods conferring a population of plants with tolerance or resistance to species of *Ca. Liberibacter* and screening that population for the plants that are tolerant or resistant to species of *Ca. Liberibacter.*

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Background/Summary

REFERENCE TO ELECTRONIC SEQUENCE LISTING

[0001] The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on May 2, 2023, is named “10457373US4_SequenceListing.xml” and is 134,000 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

[0002] The present disclosure relates to the field of biotechnology. More specifically, the disclosure relates to compositions and methods for producing plants that are resistant to *Ca. Liberibacter* infection in plants, such as Huanglongbing (HLB), also known as *Citrus* greening disease.

BACKGROUND

[0003] Currently available commercial *Citrus* plants lack tolerance or resistance to Huanglongbing (HLB), also known as *Citrus* greening disease. HLB is caused by species of the phloem-limited, gram-negative bacteria of genus *Ca. Liberibacter.* In the U.S., the predominant pathogenic species is *Ca. Liberibacter asiaticus* (Las); whereas, *Ca. Liberibacter africanus* (Laf) and *Ca. Liberibacter americanus* (Lam) are the predominant pathogenic species in South Africa and Brazil, respectively. *Ca. Liberibacter* is a vector-transmitted pathogen. The vector organisms are the Asian *Citrus* psyllid (ACP), *Diaphorina citri*, and African *Citrus* psyllid, *Trioza erytreae*. HLB was first detected in the United States in August 2005 and has rapidly moved into several *Citrus* producing areas. All commercial *Citrus* plants are susceptible to HLB, and infected *Citrus* plants will irrevocably decline. Thus, HLB has resulted in a severe decline in fruit production in Florida, where HLB has become endemic. Currently, HLB management consists of preventing trees from becoming infected, which includes protecting young flush from the *Ca. Liberibacter* vector organisms and destroying infected plant material. However, due to the lack of rapid curative methods that control HLB, new methods to prevent infection are required to stop the spread of infection and further decline of the U.S. *Citrus* industry.

SUMMARY

[0004] Certain embodiments of the disclosure relate to increasing plant resistance to infection by a bacterial species from the genus *Ca. Liberibacter.* One aspect of the present disclosure relates to modified *Citrus* plants comprising genomes in which endogenous genes or regulatory elements thereof may be modified, wherein the modification confers resistance to HLB to the modified *Citrus* plant relative to a plant of the same variety lacking the modification. The *Citrus* plant in certain embodiments may be a grapefruit tree, orange tree, sweet orange tree, or mandarin tree. Further provided are plant parts and seeds of the modified *Citrus* plant. Another aspect of the disclosure is a method of producing a commodity plant product, from the modified *Citrus* plant. In certain embodiments this method comprises collecting the commodity plant product from the modified *Citrus* plant. Further provided are commodity plant products produced by this method. In addition to modified *Citrus* plants, other plants known to be infected by *Ca. Liberibacter* such as solanaceous crops may be genomically modified to disrupted to confer resistance to such infection.

[0005] In certain embodiments, modified endogenous genes may encode any polypeptide that

interacts with any Sec-dependent effector (SDE) secreted by a bacterial species from the genus *Ca. Liberibacter.* In certain embodiments, modified regulatory elements regulate any endogenous gene that may encode any polypeptide that interacts with any SDE secreted by a bacterial species from the genus *Ca. Liberibacter.* An endogenous gene in particular embodiments may encode PP2-B2/12, Lectin, Cysteine protease, Cysteine protease 15A-like, Papain-like cysteine proteases, Myb family transcription factor, YLS9-like, Cell death suppressor protein Lls1, Acd1-Like, Acd1, accelerated cell death 2 (ACD2) protein, red chlorophyll catabolite reductase-like, NDR1/HIN1-like protein 13 (Cs8g01640), PHL5-like (Cs7g01290), and PHL5 (orange 1.1t02259). An SDE in particular embodiments may be Las4025, Las470, Las4065, Las5150, or Las4250.

[0006] Still a further aspect of the disclosure is a method of generating a modified plant comprising resistance to *Ca. Liberibacter* infection. In one embodiment, the method comprises the following steps: (a) introducing a genetic modification into the genome of a plant cell, wherein the modification is to an endogenous gene or regulatory clement thereof, wherein a polypeptide encoded by the endogenous gene interacts with an SDE secreted by a bacteria species from the genus *Ca. Liberibacter;* (b) regenerating the modified plant from the plant cell or a progenitor cell thereof, wherein the plant comprises the modification (i.e. comprises cells that possess the modification); and (c) identifying a plant comprising the modification and the resistance to *Ca Liberibacter* infection. In a specific example, the plant is a *Citrus* plant or a solanaceous crop.

[0007] In certain embodiments, modified endogenous genes may encode any polypeptide that interacts with any SDE secreted by a bacterial species from the genus *Ca. Liberibacter.* In certain embodiments, modified regulatory elements regulate any endogenous gene that may encode any polypeptide that interacts with any SDE secreted by a bacterial species from the genus *Ca. Liberibacter.* An endogenous gene in particular embodiments may encodePP2-B2/12, Lectin, Cysteine protease, Cysteine protease 15A-like, Papain-like cysteine proteases, Myb family transcription factor, YLS9-like, Cell death suppressor protein Lls1, Acd1-Like, Acd1, accelerated cell death 2 (ACD2) protein, red chlorophyll catabolite reductase-like, NDR1/HIN1-like protein 13 (Cs8g01640), PHL5-like (Cs7g01290), and PHL5 (orange1. 1102259). An SDE in particular embodiments may be Las4025, Las470, Las4065, Las5150, or Las4250. In certain embodiments, step (a) comprises a genome-editing technique. In certain embodiments, the genome-editing technique comprises a nuclease, wherein the nuclease introduces a single-strand DNA break or a double-strand DNA break. In certain embodiments, the genome-editing technique comprises a TALEN, a ZFN, meganuclease, or a CRISPR/Cas system. The disclosure still further provides a *Citrus* plant produced by this and the foregoing methods.

[0008] Still yet another aspect of the disclosure is a method for conferring a plurality of plants with a resistance to *Ca. Liberibacter* infection. In one embodiment, the method comprises the following steps: (a) introducing a genetic modification into a plurality of plants, wherein the modification is to an endogenous gene or regulatory element thereof, wherein a polypeptide encoded by the endogenous gene interacts with an SDE secreted by a bacteria species from the genus *Ca. Liberibacter;* and (b) screening the plurality of plants for the modification and a resistance to *Ca. Liberibacter* infection. The plurality of plants may include *Citrus* plants or solanaceous plants. In certain embodiments, modified endogenous genes may encode any polypeptide that interacts with any SDE secreted by a bacterial species from the genus *Ca. Liberibacter.* In certain embodiments, modified regulatory elements regulate any endogenous gene that may encode any polypeptide that interacts with any SDE secreted by a bacterial species from the genus *Ca. Liberibacter.* An endogenous gene in particular embodiments may encode PP2-B2/12, Lectin, Cysteine protease, Cysteine protease 15A-like, Papain-like cysteine proteases, Myb family transcription factor, YLS9-like, Cell death suppressor protein Lls1, Acd1-Like, Acd1, accelerated cell death 2 (ACD2) protein, red chlorophyll catabolite reductase-like, NDR1/HIN1-like protein 13 (Cs8g01640), PHL5-like (Cs7g01290), and PHL5 (orange1.1102259). An SDE in particular embodiments may be Las4025, Las470, Las4065, Las5150, or Las4250. In certain embodiments, step (a) comprises a genome-editing technique. In certain embodiments, the genome-editing technique comprises a nuclease, wherein the nuclease introduces a single-strand DNA break or a double-strand DNA break. In certain embodiments, the genome-editing technique comprises a TALEN, a ZFN, meganuclease, or a CRISPR/Cas system. The disclosure still further provides a *Citrus* plant produced by this and the foregoing methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIGS. **1**A-C show transgenic expression of CLIBASIA_04025 (Las4025) in grapefruit (*Citrus paradisi*) caused HLB-like symptoms. FIG. **1**A shows binary plasmid used for transgenic expression of CLIBASIA_04025 (Las4025) in *Citrus*. CsVMV: Cassava vein mosaic virus promoter; CaMV 35S: cauliflower mosaic virus 35S promoter; 35T: cauliflower mosaic virus terminator; NosT: nopaline synthase gene terminator; LB: left border of the T-DNA region; RB: right border of the T-DNA region; GFP: green fluorescent protein; NptII: coding sequence of neomycin phosphotransferase II; and Mature protein: nucleotide sequence encoding the mature CLIBASIA_04025 (Las4025) protein without its signal peptide. FIG. **1**B shows transgenic grapefruit plants transformed with empty vector (control) or the vector that expressed CLIBASIA_04025 (Las4025) (T1, T2, and T3). FIG. **1**C shows Western blots that confirmed transgenic grapefruit lines T1, T2, and T3 expressed CLIBASIA_04025. For these Western blots, a CLIBASIA_04025 (Las4025)-specific antibody was used; the polypeptide with SEQ ID NO:1 was the antigen used to produce this antibody.

[0010] FIG. **2** shows interactions between CLIBASIA_04025 (Las4025) and targets proteins demonstrated by yeast two-hybrid assay. Yeast cells were co-transformed with GAL4 DNA-binding domain (BD) expression vector pGBKT7 and GAL4 activation domain (AD) expression vector pGADT7. The pGBKT7 vector was either empty (EV) or contained CLIBASIA_04025 (Las4025) without its signal peptide cloned in-frame with the BD; and the pGADT7 vector contained either RLK2, Pathogenesis-related protein 10 (Cs9g03630) (PR10), or PP2-B2 (orange1.1t04174) (PP2B2) cloned in-frame with the AD. The negative control was performed with empty-AD and -BD vectors, and the positive control was performed with p53-BD and T-antigen-AD vectors. Cultures of the co-transformed yeast clones were adjusted to an OD600 of 0.2 and then spotted at three dilutions (1/10, 1/100, and 1/1000) on DDO, DDO/X, DDO/X/A, QDO, QDO/X, and QDO/X/A mediums. The images were captured after 4 days of growth. SD: synthetic defined yeast minimal medium; DDO: double dropout media, SD-Leu/-Trp; QDO: quadruple dropout media, SD/-Ade/-His/-Leu/-Trp; X: X-α-Gal; and A: Aureobasidin A.

[0011] FIGS. **3**A-B show CLIBASIA_04025 interacted with cysteine protease (Cs4g07410). FIG. **3**A shows the interaction between CLIBASIA_04025 (4025) and cysteine protease (CP, Cs4g07410) detected with a Glutathione-S-transferase (GST) pull-down assay. GST/CP was expressed in *E. coli*, immobilized on glutathione sepharose beads, and incubated with *E. coli* lysates containing Maltose-Binding Protein (MBP)-4025. The bound proteins were immunoblotted using anti-GST and anti-MBP antibodies. FIG. **3**B shows the interaction between 4025 and CP detected using Bimolecular Fluorescence Complementation (BiFC), in which EYFP fluorescence represents a protein-protein interaction. Expression vectors with 4025 without its signal peptide or CP cloned in-frame with either the N- or C-terminal fragments of EYFP (EYFPN and EYFPC) were synthesized. Citrus protoplasts were co-transformed with combinations of EYFPC or EYFPN with 4025 or CP expression vectors. The co-expression of 4025-EYFPN and EYFPC/CP and the co-expression of EYFPC-4025 and CP-EYFPN produced strong signal in the *Citrus* protoplasts; whereas, no detectable fluorescence signal was produced by the negative control combinations.

[0012] FIG. **4** shows overexpression of Myb family transcription factor (MYTL, XP_015380888.1) in Duncan grapefruit induced HLB-like symptoms. FIG. 4A shows expression of MYTL in grapefruit driven by the 35S promoter reduced plant growth. FIG. 4B shows the normal amount of starch particles, red stain, that accumulated in wildtype grapefruit. FIG. 4C shows overexpression of MYTL in grapefruit induced greater starch accumulation than displayed by wildtype grapefruits.

[0013] FIGS. 5A-E show SDE15 (identified as CLIBASIA_04025) interacts with CtACD2 protein, which negatively regulate hypersensitive reaction. FIG. 5A. Yeast-two hybrid (Y2H) assay. Full-length SDE15 fused to the GAL4 DNA binding domain (BD) was expressed in combination with full-length CtACD2 fused to the GAL4 activation domain (AD) in the yeast strain Y2HGold. Strains were grown on double dropout medium (DDO) with -Trp and -Leu and screened on quadruple dropout medium (QDO) with -Trp, -Leu, -Ade and -His supplemented with X-α-Gal and Aureobasidin A (QDO/X/A). The empty BD and AD vectors were used as the negative controls. FIG. 5B. Bimolecular fluorescence complementation (BiFC) assay. Coding sequence of SDE15 without signal peptide fused to N-terminal or C-terminal fragments of EYFP vectors pSAT6-nEYFP-C1 and pSAT6-cEYFP-C1-B were co-transformed with full-length CtCtACD2 and CtCtACD1 protein which fused to C-terminal and N-terminal fragments of EYFP vectors pSAT6-nEYFP-C1 and pSAT6-cEYFP-C1-B into *Citrus* protoplasts. The EYFP fluorescence of protoplasts were imaged 1 day after incubation using a Leica fluorescence microscope. For genuine interaction between SDE15 and CtACD2, co-expression of SDE15-EYFP.sup.N and EYFP.sup.C-CtACD2 or EYFP.sup.C-SDE15 and CtACD2-EYFP.sup.N should give strong fluorescent signals in protoplast. Co-transformations of SDE15-EYFP.sup.N and EYFP.sup.C, EYFP.sup.N and EYFP.sup.C-SDE15, SDE15-EYFP.sup.N and EYFP.sup.C-SDE15, CtACD2-EYFP.sup.N and EYFP.sup.C, EYFP.sup.N and EYFP.sup.C-CtACD2, CtACD2-EYFP.sup.N and EYFP.sup.C-CtACD2, EYFP.sup.N+EYFP.sup.C were used as negative controls which didn't produce any detectable fluorescence signal. FIG. 5C. Glutathione-S-transferase (GST) pull-down assay. GST-SDE15 and GST empty vectors were expressed in *E. coli*, immobilized on glutathione sepharose beads, and incubated with *E. coli* lysate containing MBP-CtACD2. Total cell extract (Input) and eluted protein (Elute) were immunoblotted using the anti-MBP and anti-GST antibody. FIG. 5D. Hypersensitive response (HR) assay. *Agrobacterium tumefaciens* strain GV2260 harboring binary vectors containing SDE15 and CtACD2 were infiltrated into leaves of *N. benthamiana* at the concentration of 10.sup.8 CFU ml.sup.−1. Two days later, another *Agrobacterium tumefaciens* strain GV2260 harboring the binary vector containing AvrBsT protein that can trigger HR was infiltrated on the same area of the leaves treated before. HR induction was observed and photographed 2-3 days past-inoculation. All experiments were repeated three times with the similar results, and only one leaf was presented. FIG. 5E. Electrolyte leakage associated with HR induced by AvrBsT 2 days post infiltration. Leaf discs of

[0014] AvrBsT infiltrated plants were floated on deionized water with shaking. The conductivity of the solution was measured after 4 h shaking. Error bars indicate standard error of mean (n=3). Alphabets represent significant differences in different types of samples.

[0015] FIGS. 6A-F Characterization of SDE15. FIG. 6A. Sequence analysis of SDE15. Amino acid sequence of SDE15 (96 aa) with N-terminal signal peptide (highlighted in yellow) predicted using SignalP V4.1. The cleavage site localizes between the 22.sup.nd and 23.sup.rd aa (SCG-DT). FIG. 6B. Yellowing and mottling of the leaf were observed in transgenic *Citrus* cultivar 'Duncan' plants constitutively expressing SDE15 compared with the leaf of empty-vector (EV) transgenic *Citrus*. FIG. 6C. SDE15 detection in phloem sap. Phloem sap was isolated from the bark of both healthy and HLB infected *Citrus*. T1: total bark proteins; T2: total bark proteins after phloem sap isolation; P: phloem sap. FIG. 6D. Subcellular localization of SDE15. SDE15-EYFP was co-expressed with the plasma membrane localization-marker PM-CFP or the nucleus localization-marker CFP-nucleus in leaves of *N. benthamiana*. *Agrobacterium* strains carrying the corresponding expression plasmids were infiltrated at the optical density (OD.sub.600) of 0.2. Subcellular localization of SDE15-EYFP was inspected and photographed 1 day post infiltration. Scale bars: 10 μm. FIG. 6E, FIG. **6**F. qRT-PCR analysis of SDE15 expression in different Las hosts (FIG. **6**E) and in different stages of Las infection (FIG. **6**F). Relative transcript abundances were determined using gyrase subunit A of Las (CLIBASIA_00325) and *Citrus* house keeping gene encoding glyceraldehyde-3-phosphate dehydrogenase-C (GAPDH-C) as endogenous controls. Bars represent the mean of eight replicates. Asterisks represent significant differences in the transcript abundance between *Citrus* and psyllids (**p-Value<0.01). Alphabets represent significant differences in samples of different Las infection stages. Error bars indicate standard error of mean (n=6). All experiments were repeated three times with the similar results.

[0016] FIGS. **7**A-B. Hypersensitive reaction (HR) was repressed in SDE15-transgenic *Citrus*. FIG. **7**A A strong HR, a form of programmed cell death (PCD), was observed in wild type Duncan grapefruit at 3 days after inoculation with *Xanthomonas citri* subsp. *citri* strain A.sup.W (XccA.sub.W). Only slight cell death was observed on the XccA.sub.W-infiltrated leaves of SDE15-transgenic *Citrus* at 5 days post inoculation. XccA.sup.W cells were infiltrated into *Citrus* leaves at a concentration of 10.sup.8 CFU/ml. FIG. **7**B. qRT-PCR analysis of PR genes. Expression of PR1, PR3 and PR5 was repressed in SDE15-transgenic *Citrus* compared to that in wild type Duncan after HR induction by XccA.sup.W. The house keeping gene encoding glyceraldehyde-3-phosphate dehydrogenase-C (GAPDH-C) was used as an endogenous control. Bars represent the mean of four replicates. Error bars indicate standard error of mean. All experiments were repeated three times with the similar results.

[0017] FIG. **8**. Transgenic SDE15 *Citrus* plants are more susceptible to HLB. Leaf images taken 3 months post HLB infection via budding grafting. The Las titer in SDE15-transgenic *Citrus* and EV-transgenic control *Citrus* were determined by TaqMan qPCR 0, 1, 2, and 3 months post HLB infection. Each Ct value was represented by Means±standard error (n.sub.SDE15=7, n.sub.EV=5). Asterisks represent significant differences in the Las titer between SDE15-transgenic *Citrus* and non-transgenic control (**p-Value<0.01).

[0018] FIGS. **9**A-B SDE15-transgenic *Citrus* became more susceptible to *Citrus* canker caused by a virulent strain (XacA 306) of *Xanthomonas citri* pv. *citri*. FIG. **9**A. Water-soak symptom (grey color) was observed on SDE15-transgenic *Citrus* at 5 days post XacA 306 inoculation. FIG. **9**B. Bacterial population increase of XacA 306 in SDE15-transgenic *Citrus* was faster than in non-transgenic Duncan grapefruit. Bacterial cells were infiltrated into *Citrus* leaves at a concentration of 10.sup.6 CFU/ml. Error bars indicate standard error of mean (n=4). Asterisks represent significant differences in the bacteria population between SDE15-transgenic *Citrus* and non-transgenic control (**p-Value<0.01, ***p-Value<0.001).

[0019] FIGS. **10**A-E. SDE15 interacts with CsACD2 protein. FIG. **10**A. Yeast-two hybrid (Y2H) assay using SDE15 as the bait and full-length CsACD2 protein as prey. Full-length SDE15 fused to the GAL4 DNA binding domain (BD) was expressed in combination with full-length CsACD2 fused to the GAL4 activation domain (AD) in the yeast strain Y2HGold. Strains were grown on double dropout medium (DDO) with -Trp and -Leu and screened on quadruple dropout medium (QDO) with -Trp, -Leu, -Ade and -His supplemented with X-α-Gal and Aurcobasidin A (QDO/X/A). The empty BD and AD vectors were used as the negative controls. FIG. **10**B Bimolecular fluorescence complementation (BiFC) assay. The coding sequence of SDE15 (without its signal peptide) fused to that of the N-terminal or C-terminal fragment of EYFP in vectors pSAT6-nEYFP-C1 and pSAT6-cEYFP-C1-B, respectively, was co-transformed into *Citrus* leaf protoplasts with full-length CsACD2 or CsACD2 protein which fused to the C-terminal or N-terminal fragment of EYFP in vectors pSAT6-nEYFP-C1 and pSAT6-cEYFP-C1-B. The EYFP fluorescence of protoplasts were imaged, 1 day after incubation, using a Leica fluorescence microscope. Co-transformations of SDE15-EYFP.sup.N and EYFP.sup.C, EYFP.sup.N and EYFP.sup.C-SDE15, SDE15-EYFP.sup.N and EYFP.sup.C-SDE15, CSACD2-EYFP.sup.N and EYFP.sup.C, EYFP.sup.N and EYFP.sup.C-CsACD2, CSACD2-EYFP.sup.N and EYFP.sup.C-CsACD2, EYFP.sup.N+EYFP.sup.C were used as negative controls, which did not produce any detectable fluorescence signal. FIG. **10**C, FIG. **10**D, FIG. **10**E. Glutathione-S-transferase (GST) pull-down assay. GST-SDE15 and GST empty vectors were expressed in *E. coli*, immobilized on glutathione sepharose beads, and incubated with *E. coli* lysate containing MBP-CsACD2. Total cell extract (Input) and eluted protein (Elute) were immunoblotted using the anti-MBP and anti-GST antibody.

[0020] FIG. **11**. Subcellular localization of CsACD2 and co-localization of SDE15 and CsACD2. CsACD2-EYFP was co-expressed with the plasma membrane localization-marker PM-CFP, the nucleus-marker CFP-nucleus or SDE15-CFP in leaves of *N. benthamiana. A. tumefaciens* strain GV2260 harboring the corresponding plasmids were infiltrated into leaves at OD.sub.600 of 0.2. Subcellular localization was inspected and photographed 1 day post infiltration. Scale bars: 10 μm.

[0021] FIGS. **12**A-C. SDE15 represses the hypersensitive response in tobacco and promotes the RCCR activity of CsACD2 in vitro. FIG. **12**A. Hypersensitive response (HR) assay. *A. tumefaciens* strain GV2260 harboring binary vectors that are designed to express SDE15, CsACD2 (Left) or truncated SDE15 (Right) were co-infiltrated into leaves of *N. benthamiana* at the concentration of 10.sup.8 CFU ml.sup.−1. Two days later, another *A. tumefaciens* strain GV2260 harboring the binary vector that is designed to express AvrBsT protein, which can trigger an HR was infiltrated on the same area of the leaves. HR induction was observed and photographed 2-3 days past-inoculation. All experiments were repeated three times with the similar results. FIG. **12**B. Electrolyte leakage associated with the HR induced by AvrBsT 2 days post infiltration. Leaf discs were floated on deionized water with shaking. The conductivity of the solution was measured after 4 h shaking. Error bars indicate standard error of mean (n=**3**). Alphabets represent significant differences in different types of samples. FIG. **12**C. Coupled PAO/RCCR assay to measure CsACD2 activity. Activity of purified recombinant CsACD2 was assessed in a coupled assay using purified PAO and co-factors. pFCC as the product was measured by HPLC. Purified GST-SDE15, SDE15.sup.ΔN or SDE15.sup.ΔC were added to the reaction mixture to examine whether full-length SDE15and truncated SDE15 proteins affect the activity of CsACD2. As negative controls, purified GST protein or mock purification of the vector alone without CsACD2 was added to the reaction system. Error bars represent SD (n=3). This experiment was done twice with similar results.

BRIEF DESCRIPTION OF THE SEQUENCES

[0022] SEQ ID NO:1 Antigen sequence used to produce CLIBASIA_04025 (Las4025)-specific antibody from *Ca. Liberibacter asiaticus.*

[0023] SEQ ID NO:2 CLIBASIA_04025 cDNA sequence from *Ca. Liberibacter asiaticus.*

[0024] SEQ ID NO:3 CLIBASIA_00470 cDNA sequence from *Ca. Liberibacter asiaticus.*

[0025] SEQ ID NO:4 CLIBASIA_04065 cDNA sequence from *Ca. Liberibacter asiaticus.*

[0026] SEQ ID NO:5 CLIBASIA_05150 cDNA sequence from *Ca. Liberibacter asiaticus.*

[0027] SEQ ID NO:6 CLIBASIA_04250 cDNA sequence from *Ca. Liberibacter asiaticus.*

[0028] SEQ ID NO:7 ACD2 Cs1g22670 gene sequence from *Citrus sinensis* cultivar Valencia.

[0029] SEQ ID NO:8 ACD2 Cs1g22670.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.

[0030] SEQ ID NO:9 AT4G37000 DNA sequence from *Arabidopsis thaliana.*

[0031] SEQ ID NO:10 AT4G37000 protein sequence from *Arabidopsis thaliana.*

[0032] SEQ ID NO:11 Cysteine Protease Cs4g07410 gene sequence from *Citrus sinensis* cultivar Valencia.

[0033] SEQ ID NO:12 Cysteine Protease Cs4g07410.1 cDNA variant sequence from *Citrus sinensis* cultivar Valencia.

[0034] SEQ ID NO:13 Cysteine Protease CS4g07410.2 cDNA variant sequence from *Citrus sinensis* cultivar Valencia.

[0035] SEQ ID NO:14 RCCR-like Cs1g22680 gene sequence from *Citrus sinensis* cultivar Valencia.

[0036] SEQ ID NO:15 RCCR-like Cs1g22680.1 cDNA sequence from *Citrus sinensis* cultivar

Valencia.
[0037] SEQ ID NO:16 Lls1 Cs9g02990 gene sequence from *Citrus sinensis* cultivar Valencia.
[0038] SEQ ID NO:17 Lls1 Cs9g02990.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0039] SEQ ID NO:18 ACD1-like Cs9g03000 gene sequence from *Citrus sinensis* cultivar Valencia.
[0040] SEQ ID NO:19 ACD1-like Cs9g03000.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0041] SEQ ID NO:20 ACD1 Cs8g15480 gene sequence from *Citrus sinensis* cultivar Valencia.
[0042] SEQ ID NO:21 ACD1 Cs8g15480.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0043] SEQ ID NO:22 Cysteine Proteinase 15A-like Cs3g25530 gene sequence from *Citrus sinensis* cultivar Valencia.
[0044] SEQ ID NO:23 Cysteine Proteinase 15A-like Cs3g25530.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0045] SEQ ID NO:24 Myb orange1.1t02260 gene sequence from *Citrus sinensis* cultivar Valencia.
[0046] SEQ ID NO:25 Myb orange1.1t02260.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0047] SEQ ID NO:26 YLS9-like Cs2g29120 gene sequence from *Citrus sinensis* cultivar Valencia.
[0048] SEQ ID NO:27 YLS9-like Cs2g29120.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0049] SEQ ID NO:28 Lectin orange1.1t05126 gene sequence from *Citrus sinensis* cultivar Valencia.
[0050] SEQ ID NO:29 Lectin orange1.1t05126.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0051] SEQ ID NO:30 PP2-B2/12 orange1.1t04174 gene sequence from *Citrus sinensis* cultivar Valencia.
[0052] SEQ ID NO:31 PP2-B2/12 orange1.1t04174.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0053] SEQ ID NO:32 NDR1/HIN1-like protein 13 Cs8g01640 gene sequence from *Citrus sinensis* cultivar Valencia.
[0054] SEQ ID NO:33 NDR1/HIN1-like protein 13 Cs8g01640.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0055] SEQ ID NO:34 PHL5 orange1.1t02259 gene sequence from *Citrus sinensis* cultivar Valencia.
[0056] SEQ ID NO:35 PHL5 orange1. 1t02259.1 cDNA variant sequences from *Citrus sinensis* cultivar Valencia.
[0057] SEQ ID NO:36 PHL5 orange1.1102259.2 cDNA variant sequences from *Citrus sinensis* cultivar Valencia.
[0058] SEQ ID NO:37 PHL5-like Cs7g01290 gene sequence from *Citrus sinensis* cultivar Valencia.
[0059] SEQ ID NO:38 PHL5-like Cs7g01290.1 cDNA sequence from *Citrus sinensis* cultivar Valencia.
[0060] SEQ ID NO:39 ACD2 LOC102591737 gene sequence from *Solanum tuberosum.*
[0061] SEQ ID NO:40 ACD2 NM_001318612.1 CDS sequence from *Solanum tuberosum.*
[0062] SEQ ID NO:41 Cysteine Proteinase 3 LOC102578939 gene sequence from *Solanum tuberosum.*
[0063] SEQ ID NO:42 Cysteine Proteinase 3 XM_006342320.2 CDS sequence from *Solanum tuberosum.*
[0064] SEQ ID NO:43 Lls1 LOC102597185 gene sequence from *Solanum tuberosum* (bp 734-7789).

[0065] SEQ ID NO:44 Lls1 XM_006364374.2 CDS sequence from *Solanum tuberosum.*
[0066] SEQ ID NO:45 Lls1/ACD1-like LOC102604461 gene sequence from *Solanum tuberosum.*
[0067] SEQ ID NO:46 Lls1/ACD1-like XM_006340026.2 CDS sequence from *Solanum tuberosum.*
[0068] SEQ ID NO:47 YLS9-like LOC102602250 gene sequence from *Solanum tuberosum.*
[0069] SEQ ID NO:48 YLS9-like NM_001289011.1 CDS sequence from *Solanum tuberosum.*
[0070] SEQ ID NO:49 Myb family transcription factor LOC102578723 gene sequence from *Solanum tuberosum.*
[0071] SEQ ID NO:50 Myb family transcription factor XM_006362170.2 CDS sequence from *Solanum tuberosum.*
[0072] SEQ ID NO:51 Lls1 LOC102597185 gene sequence from *Solanum tuberosum* (bp 1-144).

DETAILED DESCRIPTION

Introduction

[0073] The disclosure provides a modified plant comprising a genetic modification to an endogenous gene or regulatory element thereof, wherein it is believed that the polypeptide encoded by said endogenous gene interacts with Sec-dependent pathway effector polypeptides secreted by pathogenic species of *Ca. Liberibacter.* The cysteine protease gene may be modified such that expression of the endogeneous gene is knocked-down or reduced, or otherwise modified such that interaction with SDE is reduced. In specific examples, the endogenous gene is cysteine protease and the modified plant is *Citrus,* wherein the *Citrus* plant exhibits increased resistance to HLB as a result of the modification. Also provided are seeds, fruit, and plant parts of such plants. In another embodiment, methods are provided for generating a modified plant that is tolerant to *Ca. Liberibacter* infection, such as *Citrus* plant that is tolerant to HLB. Methods are also provided for conferring plants with resistance to *Ca. Liberibacter* infection, such as conferring *Citrus* plants with a resistance to HLB, and screening that plurality of plants for said resistance. In specific examples this is accomplished using nucleic acid modification techniques, genome recombination techniques, genome editing techniques, or a combination thereof.

Definitions

[0074] Expression: The term “expression” as used herein refers to the transcription of a particular nucleic acid sequence to produce sense or antisense RNA or mRNA, and/or the translation of an mRNA molecule to produce a polypeptide, with or without subsequent post-translational events. Expression also encompasses production of a functional nucleic acid (e.g., an RNAi, antisense molecule, ribozyme, aptamer, etc.).
[0075] Genome editing: Modifying a genome with techniques that employ targeted mutagenesis to activate DNA repair pathways. These techniques include, but are not limited to, those that utilize endonucleases to generate single-strand and double-strand DNA breaks that activate DNA repair pathways. Genome editing techniques may also comprise systems that enable targeted editing at any genomic locus. These targeting systems include, but are not limited to, polypeptides, such as, Transcription Activator-Like Effectors (TALEs) and zinc fingers (ZFs), or nucleic acids, such as, Clustered Regularly Interspaced Short Palindromic Repeats/Cas (CRISPR/CAS) single guide RNAs or NgAgo (Argonaute) single strand DNAs. As used herein, “genome editing” and “genome-engineering” are interchangeable.
[0076] Genetic modification: A DNA sequence difference, epigenetic difference, or combination thereof between two genomes of the same species in which one genome is identified as the modified genome and the other is identified as the unmodified genome and the DNA sequence or epigenetic difference is the result of applying genome modifying techniques to the unmodified genome to yield the modified genome. A genetic modification, as used herein, encompasses any insertion, deletion, or substitution of a nucleotide sequence of any size and nucleotide content, any epigenetic modification to any number of nucleotides, or a combination thereof. A genetic modification, as used herein, may also encompass introduction of one or more exogenous coding

nucleic acids that do not integrate into the unmodified genome, yet are capable of autonomous replication. In certain embodiments, a modification to an endogenous gene or regulatory element thereof may be a deletion, a substitution, or an insertion that reduces expression of the endogenous gene or the polypeptide for which it encodes. In specific embodiments, the modification may be an indel, wherein the indel may cause a frameshift mutation, a missense mutation, a nonsense mutation, a neutral mutation, or a silent mutation. In specific embodiments, a modification to a regulatory element of an endogenous gene may alter or eliminate a function of the regulatory clement. In further contemplated embodiments, the modification may comprise a nucleic acid sequence that provides exogenous control of endogenous gene, mRNA, or polypeptide expression levels. In specific embodiments, the modification may also disrupt a post-translational process of a polypeptide encoded by an endogenous gene. Post-translational processes in certain embodiments may be post-translational modification, protein sorting, or proteasomal degradation.

[0077] Genetically modified cell: A cell in which the endogenous genome has been genetically modified; a cell in which one or more exogenous, coding nucleic acids have been introduced that do not integrate into the genome, yet are capable of autonomous replication; or a combination thereof.

[0078] Genetically modified plant: A plant comprising at least one genetically modified cell. A genetically modified plant may be regenerated from a genetically modified cell or plant part comprising genetically modified cells, and thus the genetic modification may be heritable and inherited by progeny thereof. The progeny thereof that inherit the genetic modification are also considered genetically modified plants. A genetically modified plant, as used herein, also refers to a plant in which at least one genetically modified cell is introduced to a plant or arises as a result of genetic modification techniques directly applied to the plant.

[0079] Genetic modification techniques: Any technique known to those in the art that can modify the genome of a cell including, but not limited to, genome editing, site-specific genetic recombination, epigenetic modifications, and genetic transformation.

[0080] Genetic Transformation: A process of introducing a DNA sequence or construct (e.g., a vector or expression cassette) into a cell or protoplast in which that exogenous DNA is incorporated into a chromosome or is capable of autonomous replication.

[0081] Heterologous: A sequence which is not normally present in a given host genome in the genetic context in which the sequence is currently found. In this respect, the sequence may be from another species, organism, plant, tree, or variety, or may be native to the host genome, but be rearranged with respect to other genetic sequences within the host sequence. For example, a regulatory sequence may be heterologous in that it is linked to a different coding sequence relative to the native regulatory sequence. In addition, a particular recombinant DNA molecule may be heterologous with respect to a cell or organism into which it is inserted when it would not naturally occur in that particular cell or organism.

[0082] Overexpress: As used herein, “overexpress” refers to increased expression of a gene or coding sequence over that found in nature or a control plant or tissue. In some embodiments, “overexpress” may refer to greater expression of a gene or coding sequence in a genetically modified plant, when compared to a plant lacking the genetic modification.

[0083] Plant: As used herein, the term “plant” refers to *Citrus* or solanaceous plant, or any other plant that can be infected by a *Ca Liberibacter* species.

[0084] Plant part: The term “plant part” refer to cells, tissues, organs, seeds, and severed parts (e.g., roots, leaves, and flowers) that retain the distinguishing characteristics of the parent plant. “Seed” refers to any plant structure that is formed by continued differentiation of the ovule of the plant, following its normal maturation point at flower opening, irrespective of whether it is formed in the presence or absence of fertilization and irrespective of whether or not the seed structure is fertile or infertile. A plant part may be any part of the plant from which another plant may arise.

[0085] Promoter: A recognition site on a DNA sequence or group of DNA sequences that provides an expression control element for a structural gene and to which RNA polymerase specifically binds and initiates RNA synthesis (transcription) of that gene.

[0086] R.sub.0 genetically modified plant: A plant that has been genetically modified or has been regenerated from a plant cell or cells that have been genetically modified.

[0087] Reduction of Expression: The term "Reduc(e), (es) or (ing) the expression" of a gene or polypeptide in a plant or a plant cell includes inhibiting, interrupting, knocking-out, or knocking-down the gene or polypeptide, such that transcription of the gene and/or translation of the encoded polypeptide is reduced as compared to a corresponding control plant, plant cell, or population of plants or plant cells in which expression of the gene or polypeptide is not inhibited, interrupted, knocked-out, or knocked-down. "Reduced expression" encompasses any decrease in expression level (e.g., a decrease of 10% or more, 20% or more, 30% or more, 40% or more, 50% or more, 60% or more, 70% or more, 80% or more, 90% or more, or even 100%) as compared to the corresponding control plant, plant cell, or population of plants or plant cells. In some embodiments, reducing expression by 50% or more may be particularly useful. Expression levels can be measured using methods such as, for example, reverse transcription-polymerase chain reaction (RT-PCR), Northern blotting, dot-blot hybridization, in situ hybridization, nuclear run-on and/or nuclear run-off, RNase protection, or immunological and enzymatic methods such as ELISA, radioimmunoassay, and western blotting

[0088] Regeneration: The process of growing a plant from a plant cell (e.g., plant protoplast, callus, or explant).

[0089] Rootstock: As used herein, a "rootstock" refers to underground plant parts such as roots, from which new above-ground growth of a plant or tree can be produced. In accordance with the disclosure, a rootstock may be used to grow a different variety through asexual propagation or reproduction such as grafting. As used herein, a "scion" refers to a plant part that is grafted onto a rootstock variety. A scion may be from the same or a different plant type or variety.

[0090] Site-specific genome modification: Any genome modification technique that employs an enzyme that can modify a nucleotide sequence in a sequence-specific manner. Site-specific genome modification enzymes include, but are not limited to, nucleases, endonucleases, recombinases, invertases, transposases, methytransferases, demethlylases, aminases, deaminases, helicases, and any combination thereof.

[0091] Transformation construct: A chimeric DNA molecule which is designed for introduction into a host cell by genetic transformation. Preferred transformation constructs will comprise all of the genetic elements necessary to direct the expression of one or more exogenous nucleic acid sequences. In particular embodiments of the instant disclosure, it may be desirable to introduce a transformation construct into a host cell in the form of an expression cassette.

[0092] Transgene: A segment of DNA which has been incorporated into a host genome or is capable of autonomous replication in a host cell and is capable of causing the expression of one or more nucleic acid sequences. Exemplary transgenes will provide the host cell, or plants regenerated therefrom, with a novel phenotype relative to the corresponding non-transformed cell or plant. Transgenes may be directly introduced into a plant by genetic transformation, or may be inherited from a plant of any previous generation which was modified with the DNA segment.

[0093] Vector: A DNA molecule designed for transformation into a host cell. Some vectors may be capable of replication in a host cell. A plasmid is an exemplary vector, as are expression cassettes isolated therefrom.

[0094] Tolerance or resistance: Tolerance encompasses any relief from, reduced presentation of, improvement of, or any combination thereof of any symptom of an infection by a *Ca. Liberibacter* species,. Resistance encompasses tolerance as well as a reduction of bacteria upon infection or reduction of ability to infect by a *Ca. Liberibacter* species. In specific embodiments of the disclosure, *Citrus* plant may be provided that are defined as comprising a complete or less than complete resistance or tolerance to HLB. This may be assessed, for example, relative to a *Citrus*

plant not comprising a genetic modification according to the disclosure.
[0095] Hypersensitive Response (or Reaction): The hypersensitive response (or sometimes referred to a hypersensitive reaction) (HR) is plant defense mechanism that protects a plant against infection by a plant pathogen. HR is a form of cell death often associated with plant resistance to pathogen infection to prevent the spread of the potential pathogen from infected to uninfected tissues. Cell death is activated by recognition of pathogen-derived molecules by the resistance (R) gene products, and is associated with the massive accumulation of reactive oxygen species (ROS), salicylic acid (SA), and other pro-death signals such as nitric oxide (NO). *Ca. Liberibacter* species inhibit hypersensitive response, which inhibits the plant from defending itself against the *Ca. Liberibacter, Xanthomonas* species, and other pathogens It is shown herein that secretion of SDEs by a bacterial species inhibit HR. The genomic modifications described herein prevent or minimize inhibition of HR by SDES.
Detailed Description of Embodiments
[0096] The disclosure provides a significant improvement over the art due to the lack of agronomically acceptable *Citrus* plants with tolerance or resistance to HLB. HLB is a disease caused by species of the phloem-limited, gram-negative bacteria of genus *Ca. Liberibacter.* In the U.S., the predominant pathogenic species is *Ca. Liberibacter asiaticus* (Las); whereas *Ca. Liberibacter africanus* (Laf) and *Ca. Liberibacter americanus* (Lam) are the predominant pathogenic species in South Africa and Brazil, respectively. *Ca. Liberibacter* is a vector-transmitted pathogen. The vector organisms are the Asian *Citrus* psyllid, *Diaphorina citri,* and African *Citrus* psyllid, *Trioza erytreae.* HLB was first detected in the United States in August 2005 and has rapidly moved into several *Citrus* producing areas. All commercial *Citrus* plants are susceptible to HLB, and infected *Citrus* plants will irrevocably decline. Plant decline is usually preceded by a decline in the quality of the fruit and fruit drop. Fruit from infected plants are smaller, yield less juice, and have higher acidity, lower sugar and greener peel color than those from uninfected plants.
[0097] HLB has resulted in a severe decline in fruit production in Florida, where it has become endemic. However, due to the lack of rapid curative methods that control HLB, prevention of new infections is essential in HLB management. Currently, HLB management consists of preventing trees from becoming infected, which includes protecting young flush from HLB vector organisms and destroying infected plant material.
[0098] New infections could be prevented, and the disease could be managed, by planting trees that are tolerant or resistant to the disease. However, utilization of resistant germplasm to slow the spread of HLB is difficult due to the lack of commercially available resistant rootstock/scion combinations. Identification and incorporation of resistance traits from tolerant *Citrus* species and relatives is also a potential disease management strategy, but applying conventional plant breeding methods to *Citrus* plants is difficult and time consuming due to their level of nucellar embryony and long juvenile phases.
[0099] Genetically modifying *Citrus* plants is a viable alternative to conventional plant breeding. It is a relatively rapid process and some techniques allow for targeted modification of genetic locus without significant off-target effects. In such cases, genetic modification of existing cultivars has been a key component to combat HLB. In some embodiments, the disclosure employs genetic modification to render the modified *Citrus* plant tolerant to pathogenic *Ca. Liberibacter* species. In specific embodiments, the disclosure provides a *Citrus* plant that is tolerant to *Ca. Liberibacter* effector proteins. As will be understood to those of skill in the art, once a genetic modification conferring resistance to HLB is generated this could readily be introduced into any other cultivar by crossing.
[0100] Zebra Chip (ZC) is an economically important disease that occurs in commercial potato fields in the United States, Mexico, Central America, and New Zealand (Munyancza, J. E. Am. J. Pot Res (2012) 89: 329). ZC was first found in Mexico in 1994 then spread to the United States in 2000 (Rondon, S., Schreiber, A., Hamm, P., Olsen, N., Wenninger, E., Wohleb, C., Waters, T.,

Cooper, R., Walenta, D., and Reitz, S. 2017. Potato Psyllid Vector of Zebra Chip Disease in the Pacific Northwest. A Pacific Northwest Extension Publication. pp. 1-8.). Similar to Huanglongbing (HLB), ZC and diseases of other solanaceous crops are associated with a fastidious alpha-proteobacterium belonging to the '*Candidatus'* genus *Liberibacter*, '*Candidatus Liberibacter solanacearum'* (CLso), that is transmitted by a phloem-feeding psyllid vector, *Bactericera cockerelli* (Jagoucix, S., et al. 1994, Int. J. Syst. Bacteriol. 44:379-386.; Bové, J. M. 2006, J. Plant Pathol. 88:7-37; Pelz-Stelinski et al. 2010, J. Econ. Entomol. 103:1531-1541). CLso vectored by *B. cockerelli* results in a severe decline of potato, tomato, and pepper production. Current management of CLso consists of chemical controls using insecticides (Rondon et al. 2017). Due to the rapid spread of CLso, new methods to prevent infections are required.

[0101] Exemplary *Ca. Liberibacter* effector proteins contemplated by this disclosure are those secreted via the Sec-dependent pathway. Sec-dependent effector (SDE), as used herein, refers to any bacterial effector protein secreted from a bacterium via the Sec-dependent pathway. Pathogenic *Ca. Liberibacter* species secrete SDEs into the phloem of host plants, such as *Citrus* and solanaceous crops. As used herein, the terms "solanaceous crop" or "solanaceous plant" are used interchangeably and are directed plants of the Solanacea family including tomato (*Solanum lycopersicum* and *Solanum pennelli*); potato (*Solanum tuberosum*); eggplant (*Solanum melongena*), bell/chili peppers (*Capsicum annuum*, *Capsicum baccatum*, and *Capsicum chinense*). These SDEs interact with endogenous proteins and nucleic acids in the phloem and companion cells, disrupting normal physiology and inducing the symptoms of HLB in *Citrus.* Moreover, this same interaction can occur with SDEs related to *Ca. Liberibacter* species (e.g. *Ca. Liberibacter* solanacearum, 'CLso') infection in solanaccous crops such as tomato, pepper eggplant, tamarillo and potato (e.g. zebra chips).

[0102] Here we show that one of these secreted proteins, SDE15 also known as Las4025, targets a well-known negative regulator of plant programmed cell death (PCD) to promote infection. Las4025 could be detected in the phloem sap of Las-infected plants. Transgenic expression of Las4025 in *Citrus* promotes Las multiplication and HLB symptom development. SDE15 suppresses not only PCD induced by *Xanthomonas citri* subsp. *citri* (Xcc) in *Citrus*, but also PCD induced by AvrBsT (a PCD-eliciting *Xanthomonas* effector) in tobacco, suggesting that Las4025 is a broad-spectrum bacterial suppressor of plant PCD. Yeast two-hybrid, in vitro protein pull-down and in vivo bimolecular fluorescence complementation assays showed that SDE15 interacts with ACD2 (ACCELERATED CELL DEATH 2), a repressor of plant PCD and that it enhances the red chlorophyll catabolite reductase (RCCR) activity of ACD2 to remove porphyrin-related molecules, accumulation of which causes PCD. Las4025 promotes the chlorophyll break-down in planta and contributes to the development of yellowing symptom associated with HLB. Characterization of Las4025 unravels an elusive aspect of the mechanism of a major plant disease.

[0103] In some embodiments, a modified plant no longer expresses endogenous molecules, for example, polypeptides and nucleic acids, capable of interacting with *Ca. Liberibacter* SDEs. In specific embodiments, the SDEs are those secreted by Las. In more specific embodiments, an Las SDE is selected from CLIBASIA_04025 (Las4025), CLIBASIA_00470 (Las470), CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250), which are encoded by the cDNA sequences corresponding to SEQ ID NOs: 2-6.

[0104] In other embodiments, a modified plant no longer expresses an endogenous molecule, for example, a polypeptide or nucleic acid, and which may be capable of interacting with a *Ca. Liberibacter* Sec-dependent effector (SDE), that is modified. A susceptibility protein or S-protein, as used herein, refers to an endogenous host polypeptide targeted by an SDE. A susceptibility gene or S-gene, as used herein, refers to an endogenous host gene encoding an S-protein. An S-protein-SDE complex, as used herein refers to an S-protein interacting with an SDE. An S-protein-SDE interaction, as used herein refers to a protein-protein interaction between an S-protein and an SDE. In some embodiments, an S-gene is modified such that the encoded S-protein is no longer capable

of interacting with an SDE. In other embodiments, an S-gene is modified such that the encoded S-protein may interact with an SDE, but not disrupt normal physiology to an extent that a deleterious mechanism of action is triggered, for a non-limiting example, a modified S-gene that promotes proteasomal degradation of an SDE-S-protein complex before the complex activates a deleterious mechanism of action. In specific embodiments, an S-protein is selected from the group consisting of [accession numbers for *Citrus* provided in parentheses for each S-protein group] PP2-B2/12 (orange1.1t04174), Lectin (orange1.1t05126), Cysteine protease (Cs4g07410), Cysteine protease 15A-like (Cs3g25530), Papain-like cysteine proteases, Myb family transcription factor (orange1.1t02260), YLS9-like (Cs2g29120), Cell death suppressor protein Lls1 (Cs9g02990.1), Acd1-Like Cs9g03000, Acd1 Cs8g15480, accelerated cell death 2 (ACD2) protein (AT4G37000.1, Cs1g22670), red chlorophyll catabolite reductase-like (Cs1g22680), NDR1/HIN1-like protein 13 (Cs8g01640), PHL.5-like (Cs7g01290), and PHL5(orange 1.1t02259), for which cDNA examples of the *Citrus* versions are encoded by the cDNA sequences corresponding to SEQ ID NOs: 7-29 and SEQ ID NO:31. Potato orthologs are encoded by SEQ ID Nos 39-51. Provided below in Table 1 are accession numbers for select *Citrus* S-protein sequences and S-genes encoding such S-proteins, as well as orthologs in solanaccous plants, that may be modified as taught herein (Cs or orange1=gene id (*Citrus* genome database citrusgenomedb.org); NC or NW=genome sequence (NCBI database); XM=cDNA accession no. (NCBI database); LOC=gene accession no (NCBI database); and XP, NP, PHT or PHU=polypeptide accession no. (NCBI database)):
TABLE-US-00001 TABLE 1 Cell death suppressor Species Myb family transcription protein Lls1 ACD2 Lectin *Citrus sinensis* 1. orange1.1t02260 1. Cs9g02990.1 1. Cs1g22680 1. orange1.1t05126 LOC102621262 LOC102615553 XM_006466545 LOC102630138 XM_025093288 XM_006488849 LOC102623285 XM_006495169 XP_024949056 XP_006488912 XP_006466608 XP_006495232 NW_006257094.1 NC_023054.1 NC_023046.1 NW_006257465.1 (602674 . . . 604650) (1382910 . . . 1386383) (25356426 . . . 25358668) (8465 . . . 9844) 2. XP_024949055 2. Cs9g03000 2. Cs1g22670 2. LOC107177625 3. orange1.1t02259 XM_006488848 NC_023046.1 XM_015531689 LOC102608693 LOC102615272 XM_006466544 XP_015387175 XM_015525401 XP_006488911 (LOC102622999) 3. XP_015387176 XP_015380887 NC_023054.1 XP_006466607 LOC107177626 NW_006257094.1 (1390097 . . . 1395026) NC_023046.1 4. XP_015387172 (597073 . . . 599506) 3. Cs8g15480, (25352463 . . . 25354069) LOC107177622 4. LOC102608059 XM_006487933 6. XP_006475932 XM_015531655 XP_006487996 LOC102628131 XP_015387141 NC_023053.1 7. XP_015387174, 5.XP_015380888.1 (18675482 . . . 18679902) LOC107177624 *Capsicum* LOC107843940, PHT80565, LOC107868112, PHT71355, *annuum* XP_016543872, XP_016571811, PHT83236, PHT71353 XP_016543873, XP_016571812, XP_16570190, XP_016576061, XP_016571813 NP_001311893, PHT77744, XP_016557361 XP_016576060, XP_016565589, PHT82561, XP_016565591, XP_016573871, XP_016544890, PHT71362 *Capsicum* PHT37283, PHT54548, PHT50387, PHT37058 *baccatum* PHT44431, PHT54549, PHT58680 PHT52438, PHT33802, PHT48923, PHT33064, PHT33349, PHT45532 PHT60401, PHT39991 *Capsicum* PHU06047, PHU16680, PHU19541 PHT99312, *chinense* PHU13451, PHU16678, PHU_05788 PHU22243, PHU16679, PHU18686, PHU03790, PHU05798 PHU02689, PHU01515 *Solanum* 1. LOC101251632 1. LOC101255583, 1. LOC778267 N/a *lycoperiscum* NC_015447.3 NC_015441.3 NC_015440.3 (60577349 . . . 60580659 (11974361 . . . 11979604) (9353792 . . . 9357403) complement) XP_004237332, AAL32300, NP_001234535 *Solanum* LOC107032497(XP_015089588), XP_015073606, LOC107014711, N/a *pennelli* XP_015088029, XP_015058211, XP_015070234 XP_015088022, XP_015072446 XP_015078237, XP_015078236, XP_015076100, XP_015072629, XP_015055275 *Solanum* 1. LOC102578723 1. LOC102597185 1. LOC102591737, N/a *tuberosum* NW_006239309.1 NW_006239415.1 NW_006239292.1 (222724 . . . 226103) (360276 . . . 368064) (123266 . . . 127905 2. LOC102604461 complement) NW_006238942.1 NP_001305541 (19429 . . . 23712) Species Cysteine protease PP2-B12 YLS9-

like *Citrus sinensis* 1. Cs4g07410 1. orange1.1t04174 1. Cs2g29120 LOC102578016 LOC102626181 LOC102624273 XM_006474664 XM_025094054 XM_006470378 NM_001288897 XP_024949822 XP_006470441 NP_001275826 NW_006257165.1 NC_023047.1 NC_023049.1 (68010 . . . 79424) (28676278 . . . 28677278) (4697175 . . . 4700328) 2. Cs2g29120 2. Cs3g25530 LOC107174220 XM_006473521 NC_023053.1 XP_006473584 (385414 . . . 385806 LOC102608509 complement) NC_023048.1 (27116634 . . . 27118954) *Capsicum* XP_016580127, PHT66823, XP_016552935, *annuum* XP_016557040, XP_016552209, XP_016561811, PHT84613. XP_016552365, XP_016560224, XP_016539529, PHT66822, XP_016563876, PHT90352, XP_016573353, PHT87859, XP_016561024, XP_016552297, XP_016562568, PHT70914 XP_016573817 PHT62176, PHT71162, XP_016562568 *Capsicum* PHT42963, PHT32781, PHT59305, *baccatum* PHT30454, PHT32780, PHT54760, PHT41774, PHT32035, PHT27977, PHT_57030, PHT39101, PHT53910, PHT_36437, PHT32776, PHT29394 PHT30404, PHT32782, PHT43389, PHT30386, PHT48962 PHT36895 PHT59073 *Capsicum* PHU11729, PHU01437, PHU29409, *chinense* PHU20763, PHU01440, PHU24981, PHU10453, PHU00690, PHT98447, PHU27226, PHU21394, PHU24513, PHU05443 PHU01435, PHU12324, PHU18677, BAD11071 PHT98988, PHU10308 *Solanum* 1. LOC101252505 XP_004239687, 1. LOC101250915 *lycoperiscum* NC_015444.3 XP_004253004, NC_015438.3 (54626241 . . . 54628525 XP_004237494, (3106210 . . . 3109693 complement) XP_004237583, complement) XP_010314855, XP_004252380 *Solanum* XP_015082349, XP_015074926, XP_015084054, *pennelli* XP_015081027, XP_015071726, XP_015086729, XP_015074247, XP_015059542, XP_015065001, XP_015068628, XP_015073190, XP_015067199, XP_015061093, XP_01505954, XP_015070124, XP_015058018, XP_015084179, XP_015081836 XP_015063485, XP_015060715 XP_015069437 *Solanum* 1. LOC102578939 XP_006349935, 1. LOC102602250 *tuberosum* NW_006238961.1 XP_006345814, NW_006238997.1 (2218619 . . . 2220963 XP_006340500, (685210 . . . 686001 complement) XP_006366161, complement) XP_015165222, XP_006344703, XP_006361502, XP_006351708

[0105] This disclosure also contemplates embodiments in which a genetic modification anywhere in the genome disrupts expression, and in turn, may disrupt an S-protein-SDE interaction from activating a deleterious mechanism of action. In some embodiments, an S-gene or a regulatory element thereof is modified. In some embodiments, a modification may be made elsewhere in the genome, but ultimately disrupts an S-protein-SDE interaction, for a non-limiting example, inserting genetic material encoding a polypeptide capable of disrupting an S-protein-SDE interaction anywhere in the genome. In specific embodiments, an SDE is secreted the *Ca. Liberibacter* species selected from the group consisting of Las, Laf, and Lam. In some specific embodiments, the SDEs are CLIBASIA_0402 (Las4025), CLIBASIA_00470 (Las470), CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250). In specific embodiments, an S-protein is selected from the group consisting of PP2-B2/12 (orange 1.1t04174), Lectin (orange1.1t05126),, Cysteine protease (Cs4g07410), Cysteine protease 15A-like (Cs3g25530), Papain-like cysteine proteases, Myb family transcription factor (orange 1.1t02260), YLS9-like (Cs2g29120),, Cell death suppressor protein Lls 1 (Cs9g02990.1), Acd1-Like Cs9g03000, Acd1 Cs8g15480, accelerated cell death 2 (ACD2) protein (AT4G37000.1, Cs1g22670), red chlorophyll catabolite reductase-like (Cs1g22680), NDR1/HIN1-like protein 13 (Cs8g01640), PHL5-like (Cs7g01290), and PHL5 (orange1. It02259).

[0106] In some cases, a modification is conducted at a target sequence as set forth in Table 1, or at a target sequence that is at least 95 percent (e.g., at least 96 percent, at least 97 percent, at least 98 percent, or at least 99 percent) identical to the sequence set forth in Table 1. In a more specific example, a modification is conducted at a target sequence set forth in SEQ ID Nos 7-38 or 39-51, or at a target sequence that is at least 95 percent (e.g., at least 96 percent, at least 97 percent, at least 98 percent, or at least 99 percent) identical to a sequence set forth in SEQ ID Nos 7-38 or 39-51.

[0107] The percent sequence identity between a particular nucleic acid or amino acid sequence and a sequence referenced by a particular sequence identification number may be determined by techniques known in the art. In one example, sequence identity is determined as follows. First, a nucleic acid or amino acid sequence is compared to the sequence set forth in a particular sequence identification number using the BLAST 2 Sequences (Bl2seq) program from the stand-alone version of BLASTZ containing BLASTN version 2.0.14 and BLASTP version 2.0.14. This stand-alone version of BLASTZ can be obtained online at fr.com/blast or at ncbi.nlm.nih.gov. Instructions explaining how to use the Bl2 seq program can be found in the readme file accompanying BLASTZ. Bl2seq performs a comparison between two sequences using either the BLASTN or BLASTP algorithm. BLASTN is used to compare nucleic acid sequences, while BLASTP is used to compare amino acid sequences. To compare two nucleic acid sequences, the options are set as follows: -i is set to a file containing the first nucleic acid sequence to be compared (e.g., C:\seq1.txt); -j is set to a file containing the second nucleic acid sequence to be compared (e.g., C:\seq2.txt); -p is set to blastn; -o is set to any desired file name (e.g., C:\output.txt); -q is set to −1; -r is set to 2; and all other options are left at their default setting. For example, the following command can be used to generate an output file containing a comparison between two sequences: C:\Bl2seq -i c:\seq1.txt -j c:\seq2.txt -p blastn -o c:\output.txt -q −1 -r 2. To compare two amino acid sequences, the options of Bl2seq are set as follows: -i is set to a file containing the first amino acid sequence to be compared (e.g., C:\seq1.txt); -j is set to a file containing the second amino acid sequence to be compared (e.g., C:\seq2.txt); -p is set to blastp; -o is set to any desired file name (e.g., C:\output.txt); and all other options are left at their default setting. For example, the following command can be used to generate an output file containing a comparison between two amino acid sequences: C:\Bl2seq -i c:\seq2.txt -j c:\seq2.txt -p blastp -o c:\output.txt. If the two compared sequences share homology, then the designated output file will present those regions of homology as aligned sequences. If the two compared sequences do not share homology, then the designated output file will not present aligned sequences.

[0108] Once aligned, the number of matches is determined by counting the number of positions where an identical nucleotide or amino acid residue is presented in both sequences. The percent sequence identity is determined by dividing the number of matches either by the length of the sequence set forth in the identified sequence (e.g., SEQ ID NO:8), or by an articulated length (e.g., 100 consecutive nucleotides or amino acid residues from a sequence set forth in an identified sequence), followed by multiplying the resulting value by 100. For example, a nucleic acid sequence that has 1200 matches when aligned with the sequence set forth in SEQ ID NO:8 is 83.7 percent identical to the sequence set forth in SEQ ID NO:1 (i.e., 1200÷1434×100=83.7). It is noted that the percent sequence identity value is rounded to the nearest tenth. For example, 75.11, 75.12, 75.13, and 75.14 are rounded down to 75.1, while 75.15, 75.16, 75.17, 75.18, and 75.19 are rounded up to 75.2. It also is noted that the length value will always be an integer.

[0109] The embodiments described herein are not limited to a particular *Citrus* or solanaceous plant or variety but rather encompass any *Citrus* or solanaceous plant or hybrid thereof that may be useful in accordance with the disclosure. Citrus varieties contemplated by this disclosure include, but are not limited to, cultivated *Citrus* types such as sweet orange, bitter orange, blood orange, grapefruit, pomelo, citron, clementine, naval orange, lemon, lime, mandarin, tangerine, tangelo, or the like.

I. Genome Editing

[0110] Certain aspects of the present disclosure relate to methods of modifying the genome of a *Citrus* or solanaceous plant using genome editing techniques. As used herein, “genome editing” and “genome-engineering” are terms used interchangeably and refer to the modification of a genome through mutagenesis. For example, in plant genome engineering, endonucleases may be used to generate double-strand DNA breaks (DSBs) and activate genome repair pathways. These DSB repair pathways may repair the break cleanly, i.e., without altering the starting sequence, or, alternatively, induce a mutation through an error in repair. In some embodiments, genome editing is used to insert, delete, or substitute one or more base pairs at one or any combination of genetic loci. In some embodiments, a genome editing technique is used to create a mutation, for example, a point mutation or single nucleotide polymorphism.

[0111] In some embodiments the DSB repair pathway is non-homologous end-joining (NHEJ) or microhomology mediated end joining (MMEJ). During NHEJ, any nucleotide overhangs on the break ends are either resected or filled to form blunt ends that are ligated. During MMEJ, the break ends are processed to reveal overhangs comprising microhomology sequences that are then ligated together. The insertions or deletions resulting from the terminal end processing in both the NHEJ and MMEJ pathways can be referred to as indels. In some embodiments, the NHEJ or MHEJ that occurs can be relied upon to introduce a genome modification including, but not limited to, a silent mutation, a neutral mutation, a missense mutation, a nonsense mutation, or a frameshift mutation.

[0112] In other embodiments, the DSB repair pathway is homologous recombination (HR). During HR, a DSB is repaired using a template with sequences with homology to the DNA flanking the break, i.e., a homologous chromosome. In plant genome editing, a linear DNA polynucleotide flanked by sequences (e.g., of 50 base pairs or more) homologous to those flanking a targeted genomic locus, may be introduced into the genome when a DSB is repaired by HR. In some embodiments, this approach is used to introduce, substitute, or delete a DNA sequence at a genomic locus. Any DNA sequence of interest may be introduced, deleted, or substituted. An introduced or substituted DNA sequence may encode an RNA molecule with a specific activity or function, a DNA molecule with a specific activity or function (e.g., encoding a polypeptide, representing a detectable marker, etc.), a DNA molecule comprising cis-regulatory elements, or a DNA molecule encoding a polypeptide, a motif thereof, or domain thereof. In some embodiments, the nucleic acid encoding the linear DNA sequence that will act as the HR template is encoded by an expression vector. In some embodiments, the nucleic acid encoding the linear DNA sequence of interest is encoded by a DNA sequence separate from the expression vector. For example, and without limitation, the nucleic acid encoding a DNA sequence of interest may be a linear DNA polynucleotide that is co-transformed with an expression vector.

[0113] In some embodiments, single-strand breaks or “nicks” are introduced into the target DNA sequence. As used herein, the term “single-strand break inducing agent” or “nickase” refers to any agent that can induce a single-strand break (SSB) in a DNA molecule. In some embodiments two SSBs are introduced into the target DNA to generate a DSB. These breaks may also be repaired by HR, NHEJ, or MMEJ. In some embodiments, sequence modifications occur at or near the SSB sites, which can include deletions or insertions that result in modification of the nucleic acid sequence, or integration of exogenous nucleic acids by HR or NHEJ.

[0114] In one aspect, a “modification” comprises the insertion of at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 25, at least 50, at least 100, at least 200, at least 300, at least 400, at least 500, at least 750, at least 1000, at least 1500, at least 2000, at least 3000, at least 4000, at least 5000, or at least 10,000 nucleotides. In another aspect, a “modification” comprises the deletion of at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 25, at least 50, at least 100, at least 200, at least 300, at least 400, at least 500, at least 750, at least 1000, at least 1500, at least 2000, at least 3000, at least 4000, at least 5000, or at least 10,000 nucleotides. In a further aspect, a “modification” comprises the inversion of at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 25, at least 50, at least 100, at least 200, at least 300, at least 400, at least 500, at least 750, at least 1000, at least 1500, at least 2000, at least 3000, at least 4000, at least 5000, or at least 10,000 nucleotides. In still another aspect, a “modification” comprises the substitution of at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 15, at least 25, at least 50, at least 100, at least 200, at least 300, at least 400, at least 500, at least 750, at least 1000, at least 1500, at least 2000, at least 3000, at least 4000, at least 5000, or at least 10,000 nucleotides. In some embodiments, a "modification" comprises the substitution of an "A" for a "C," "G" or "T" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of an "C" for an "A," "G" or "T" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of a "G" for an "A," "C" or "T" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of a "T" for an "A," "C" or "G" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of a "C" for an "U" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of a "G" for an "A" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of an "A" for a "G" in a nucleic acid sequence. In some embodiments, a "modification" comprises the substitution of a "T" for a "C" in a nucleic acid sequence.

[0115] In some embodiments, genome editing of a *Citrus* plant as described herein may encompass techniques that employ methods of targeting endonucleases to one or more genetic loci. In some embodiments, synthetic polypeptides, for example, Transcription Activator-Like Effectors (TALEs) and zinc fingers (ZFs), or nucleic acids, for example, Clustered Regularly Interspaced Short Palindromic Repeats/Cas (CRISPR/CAS) single guide RNAs or NgAgo (Argonaute) single strand DNAs, are used to target endonucleases to any genomic locus. The targeted endonucleases may catalyze a DSB at a target locus. Upon detecting these breaks, a cell may initiate any DSB repair pathway. In some embodiments, genome editing is carried out at more than one genomic locus simultaneously (i.e., multiplex genome engineering). In some embodiments, multiplex genome engineering may be used to remove a sequence of any size from the genome. In some embodiments, any combination and number of endonuclease targeting techniques may be used to target one or more genetic loci.

A. RNA- and DNA-Guided Genome Editing Systems

[0116] In some embodiments, genome engineering of a *Citrus* plant as described herein may employ RNA-guided endonucleases including, but not limited to CRISPR/Cas systems. CRISPR/Cas systems have been described in U.S. Patent Application Publication Nos. 2017/0191082 and 2017/0106025, each of which are incorporated herein by reference in their entirety. In some embodiments, a targeted genome modification as described herein comprises the use of at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, or at least ten RNA-guided nucleases. In some embodiments, a CRISPR/Cas9 system, a CRISPR/Cpf1 system, a CRISPR/CasX system, or a CRISPR/CasY system are alternatives that may be used to generate modifications to target sequences as described herein.

[0117] The CRISPR systems are based on RNA-guided endonucleases that use complementary base pairing to recognize DNA sequences at target sites. CRISPR/Cas systems are part of the adaptive immune system of bacteria and archaca, protecting them against invading DNA, such as viral DNA, by cleaving the foreign DNA in a sequence-dependent manner. The immunity is acquired by the integration of short fragments of the invading DNA known as spacers between two adjacent repeats at the proximal end of a CRISPR locus. The CRISPR arrays, including the spacers, are transcribed during subsequent encounters with invasive DNA and are processed into small interfering CRISPR RNAs (crRNAs) approximately 40 nt in length, which combine with the trans-activating CRISPR RNA (tracrRNA) to activate and guide the Cas9 nuclease. This cleaves homologous double-stranded DNA sequences known as protospacers in the invading DNA.

[0118] A prerequisite for cleavage is the presence of a conserved protospacer-adjacent motif (PAM) downstream of the target DNA, which usually has the sequence 5′-NGG-3′ but less frequently NAG. Specificity is provided by the so-called "seed sequence" approximately 12 bases upstream of the PAM, which must match between the RNA and target DNA. Cpf1 acts in a similar manner to Cas9, but Cpf1 does not require a tracrRNA. Specificity of the CRISPR/Cas system is based on an RNA-guide that use complementary base pairing to recognize target DNA sequences. In some embodiments, the site-specific genome modification enzyme is a CRISPR/Cas system. In an aspect, a site-specific genome modification enzyme provided herein can comprise any RNA-guided Cas endonuclease (non-limiting examples of RNA-guided nucleases include Cas1, Cas1B, Cas2, Cas3, Cas4, Cas5, Cas6, Cas7, Cas8, Cas9 (also known as Csn1 and Csx12), Cas10, Csy1,Csy2, Csy3, Cse1, Cse2, Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3,Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csx1, Csx15,Csf1, Csf2, Csf3, Csf4, Cpf1, homologs thereof, or modified versions thereof); and, optionally, the guide RNA necessary for targeting the respective nucleases.

[0119] In some embodiments, an RNA-guided endonuclease is the DNA cleavage domain of a restriction enzyme fused to a deactivated Cas9 (dCas9), for example dCas9-Fok1. As used herein, a "dCas9" refers to a endonuclease protein with one or more amino acid mutations that result in a Cas9 protein without endonuclease activity, but retaining RNA-guided site-specific DNA binding. As used herein, a "dCas9-restriction enzyme fusion protein" is a dCas9 with a protein fused to the dCas9 in such a manner that the restriction enzyme is catalytically active on the DNA.

[0120] In some embodiments, genome editing of a *Citrus* or solanaceous plant as described herein may employ DNA-guided endonucleases including, but not limited to, NgAgo systems.

[0121] In one aspect, a method and/or composition provided herein comprises one or more, two or more, three or more, four or more, or five or more guide RNAs or DNAs. In another aspect, a CRISPR/CAS system, dCas9-restriction enzyme fusion protein, NgAgo system provided herein is capable of generating a targeted DSB in a target sequence as described herein. In one aspect, vectors comprising nucleic acids encoding one or more, two or more, three or more, four or more, or five or more guide RNAs or DNAs and the corresponding CRISPR/CAS system, dCas9-restriction enzyme fusion protein, NgAgo system are provided to a cell by transformation methods known in the art (e.g., without being limiting, viral transfection, particle bombardment, PEG-mediated protoplast transfection or *Agrobacterium*-mediated transformation).

B. Transcription Activator-Like Effector Nucleases

[0122] In some embodiments, genome editing of a *Citrus* plant as described herein may employ Transcription Activator-Like Effector Nucleases (TALENs). TALENs have been described in U.S. Patent Application Publication Nos. 2016/0369301 and 2015/0203871 (both of which are incorporated herein by reference in their entirety) and are well known in the art. TALENs are artificial restriction enzymes generated by fusing the transcription activator-like effector (TALE) DNA binding domain to an endonuclease domain. In one aspect, the nuclease is selected from a group consisting of PvuII, MutH, TevI and FokI, AlwI, MlyI, SbfI, SdaI, StsI, CleDORF, Clo051, Pept071. The term TALEN, as used herein, is broad and includes a monomeric TALEN that can cleave double stranded DNA without assistance from another TALEN. The term TALEN is also used to refer to one or both members of a pair of TALENs that work together to cleave DNA at the same site.

[0123] TALEs can be engineered to bind practically any DNA sequence, such as a target sequence as described herein. TALE proteins are DNA-binding domains derived from various plant bacterial pathogens of the genus *Xanthomonas.* The X pathogens secrete TALEs into the host plant cell during infection. The TALE moves to the nucleus, where it recognizes and binds to a specific DNA sequence in the promoter region of a specific DNA sequence in the promoter region of a specific gene in the host genome. TALE has a central DNA-binding domain composed of 13-28 repeat monomers of 33-34 amino acids. The amino acids of each monomer are highly conserved, except for hypervariable amino acid residues at positions 12 and 13. The two variable amino acids are called repeat-variable diresidues (RVDs). The amino acid pairs NI, NG, HD, and NN of RVDs preferentially recognize adenine, thymine, cytosine, and guanine/adenine, respectively, and modulation of RVDs can recognize consecutive DNA bases. This simple relationship between amino acid sequence and DNA recognition has allowed for the engineering of specific DNA binding domains by selecting a combination of repeat segments containing the appropriate RVDs.

[0124] In one aspect, a method and/or composition provided herein comprises one or more, two or more, three or more, four or more, or five or more TALENs. In another aspect, a TALEN provided herein is capable of generating a targeted DSB in a target sequence as described herein. In one aspect, vectors comprising polynucleotides encoding one or more, two or more, three or more, four or more, or five or more TALENs are provided to a cell by transformation methods known in the art (e.g., without being limiting, viral transfection, particle bombardment, PEG-mediated protoplast transfection or *Agrobacterium*-mediated transformation).

C. Zinc Finger Nucleases

[0125] In some embodiments, genome engineering of a *Citrus* or solanaceous plant as described herein may employ Zinc Finger Nucleases (ZFNs). ZFNs have been described in U.S. Pat. No. 9,322,006 (incorporated herein by reference in its entirety) and are well known in the art. ZFNs are synthetic proteins consisting of an engineered zinc finger DNA-binding domain fused to the cleavage domain of an endonuclease, for example, Fok1. ZFNs can be designed to cleave almost any long stretch of double-stranded DNA by the modification of the zinc finger DNA-binding domain. ZFNs form dimers from monomers composed of a non-specific DNA cleavage domain of FokI nuclease fused to a zinc finger array engineered to bind a target DNA sequence. The DNA-binding domain of a ZFN is typically composed of 3-4 zinc-finger arrays. The amino acids at positions −1, +2, +3, and +6 relative to the start of the zinc finger ∞-helix, which contribute to site-specific binding to the target DNA, can be changed and customized to fit specific target sequences. The other amino acids form the consensus backbone to generate ZFNs with different sequence specificities. Rules for selecting target sequences for ZFNs are known in the art. The FokI nuclease domain requires dimerization to cleave DNA and therefore two ZFNs with their C-terminal regions are needed to bind opposite DNA strands of the cleavage site (separated by 5-7 nt). The ZFN monomer can cut the target site if the two-ZF-binding sites are palindromic. The term ZFN, as used herein, is broad and includes a monomeric ZFN that can cleave double stranded DNA without assistance from another ZFN. The term ZFN is also used to refer to one or both members of a pair of ZFNs that are engineered to work together to cleave DNA at the same site.

[0126] Without being limited by any scientific theory, because the DNA-binding specificities of zinc finger domains can in principle be re-engineered using one of various methods, customized ZFNs can theoretically be constructed to target nearly any gene sequence. Publicly available methods for engineering zinc finger domains include Context-dependent Assembly (CoDA), Oligomerized Pool Engineering (OPEN), and Modular Assembly.

[0127] Several embodiments relate to a method and/or composition provided herein comprising one or more, two or more, three or more, four or more, or five or more ZFNs directed to a target sequence as described herein. In another aspect, a ZFN provided herein is capable of generating a targeted DSB. In one aspect, vectors comprising polynucleotides encoding one or more, two or more, three or more, four or more, or five or more ZFNs are provided to a cell by transformation methods known in the art (e.g., without being limiting, viral transfection, particle bombardment, PEG-mediated protoplast transfection or *Agrobacterium*-mediated transformation).

D. Meganucleases

[0128] In some embodiments, genome engineering of a *Citrus* or solanaceous plant as described herein may employ a meganuclease. Meganucleases, which are commonly identified in microbes, are unique enzymes with high activity and long recognition sequences (>14 nt) resulting in site-specific digestion of target DNA. Engineered versions of naturally occurring meganucleases typically have extended DNA recognition sequences (for example, 14 to 40 nt). The engineering of meganucleases can be more challenging than that of ZFNs and TALENs because the DNA recognition and cleavage functions of meganucleases are intertwined in a single domain. Specialized methods of mutagenesis and high-throughput screening have been used to create novel meganuclease variants that recognize unique sequences and possess improved nuclease activity.

[0129] In one aspect, a method and/or composition provided herein comprises one or more, two or more, three or more, four or more, or five or more meganucleases directed to a target sequence as described herein. In some embodiments, a meganuclease provided herein is capable of generating a targeted DSB. In some embodiments, vectors comprising polynucleotides encoding one or more, two or more, three or more, four or more, or five or more meganucleases are provided to a cell by transformation methods known in the art (e.g., without being limiting, viral transfection, particle bombardment, PEG-mediated protoplast transfection or *Agrobacterium*-mediated transformation).

II. Site-Specific Genome Modification

[0130] Certain aspects of the present disclosure relate to methods of modifying the genome of a *Citrus* plant using site-specific genome modification techniques. In some embodiments, site-specific genome modification of a *Citrus* plant as described herein may employ any site-specific genome modification enzyme. As used herein, the term "site-specific genome modification enzyme" refers to any enzyme that can modify a nucleotide sequence in a sequence-specific manner. In some embodiments, a site-specific genome modification enzyme modifies the genome by inducing a single-strand break. In some embodiments, a site-specific genome modification enzyme modifies the genome by inducing a double-strand break. In some embodiments, a site-specific genome modification enzyme is a recombinase. In some embodiments, a site-specific genome modification enzyme is a transposase. In the present disclosure, site-specific genome modification enzymes include, but are not limited to, nucleases, endonucleases, recombinases, invertases, transposases, methytransferase, demethlylases, aminases, deaminases, helicases, and any combination thereof.

[0131] In some embodiments, the site-specific genome modification enzyme is a recombinase. Non-limiting examples of recombinases include a tyrosine and serine recombinases and coupled with a DNA recognition motifs, for example, a Cre recombinase, a Gin recombinase, a Flp recombinase, and a Tnp1 recombinase. In another aspect, a serine recombinase coupled with a DNA recognition motif, for example, a PhiC31 integrase, an R4 integrase, and a TP-901 integrase. In an aspect, a recombinase is tethered to a zinc-finger DNA-binding domain, or a TALE DNA-binding domain, or a Cas9 nuclease.

[0132] The Flp-FRT site-directed recombination system comes from the 2μ plasmid from the baker's yeast Saccharomyces cerevisiae. In this system, Flp recombinase (flippase) recombines sequences between flippase recognition target (FRT) sites. FRT sites comprise 34 nucleotides. Flp binds to the "arms" of the FRT sites (one arm is in reverse orientation) and cleaves the FRT site at either end of an intervening nucleic acid sequence. After cleavage, Flp recombines nucleic acid sequences between two FRT sites.

[0133] Cre-lox is a site-directed recombination system derived from the bacteriophage PI that is similar to the Flp-FRT recombination system. Cre-lox can be used to invert a nucleic acid sequence, delete a nucleic acid sequence, or translocate a nucleic acid sequence. In this system, Cre recombinase recombines a pair of lox nucleic acid sequences. Lox sites comprise 34 nucleotides, with the first and last 13 nucleotides (arms) being palindromic. During recombination, Cre recombinase protein binds to two lox sites on different nucleic acids and cleaves at the lox sites. The cleaved nucleic acids are spliced together (reciprocally translocated) and recombination is complete. In another aspect, a lox site provided herein is a loxP, lox 2272, loxN, lox 511, lox 5171, lox71, lox66, M2, M3, M7, or M11 site.

[0134] In another aspect, the site-specific genome modification enzyme is a dCas9-recombinase fusion protein. As used herein, a "dCas9-recombinase fusion protein" is a dCas9 with a protein fused to the dCas9 in such a manner that the recombinase is catalytically active on the DNA. In some embodiments, dCas9 may be fused with the catalytic domain of any enzyme such that the catalytic domain is catalytically active on DNA. In another aspect, a DNA transposase is attached to a DNA binding domain for example, a TALE-piggyBac and TALE-Mutator.

[0135] Several embodiments relate to promoting DNA recombination by providing a site-specific genome modification enzyme to a plant cell. In some embodiments, recombination is promoted by providing a strand separation inducing reagent. In one aspect, the site-specific genome modification enzyme is selected from an endonuclease, a recombinase, an invertase, a transposase, a helicase or any combination thereof. In some embodiments, recombination occurs between B chromosomes. In some embodiments, recombination occurs between a B chromosome and an A chromosome.

[0136] Several embodiments relate to promoting integration of one or more DNAs of interest by providing a site-specific genome modification enzyme. In some embodiments, integration of one or more DNAs of interest is promoted by providing a strand separation inducing reagent. In one aspect, the site-specific genome modification enzyme is selected from an endonuclease, a recombinase, a transposase, a helicase or any combination thereof. Any DNA sequence can be integrated into a target site of a chromosome sequence by introducing the DNA sequence and the provided site-specific genome modification enzymes. Any method provided herein can utilize any site-specific genome modification enzyme provided herein.

[0137] Several embodiments relate to a method and/or a composition provided herein comprising at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, or at least ten site-specific genome modification enzymes. In yet another aspect, a method and/or a composition provided herein comprises at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, or at least ten polynucleotides encoding at least one, at least two, at least three, at least four, at least five, at least six, at least seven, at least eight, at least nine, or at least ten site-specific genome modification enzymes.

III. Plant Transformation Constructs

[0138] Vectors used for plant transformation may include, for example, plasmids, cosmids, YACs (yeast artificial chromosomes), BACs (bacterial artificial chromosomes) or any other suitable cloning system, as well as fragments of DNA therefrom. Thus when the term “vector” or “expression vector” is used, all of the foregoing types of vectors, as well as nucleic acid sequences isolated therefrom, are included. In some embodiments, a viral vector based on a plant virus such as a *Citrus tristeza* Virus may be used in accordance with the disclosure. It is contemplated that utilization of cloning systems with large insert capacities will allow introduction of large genetic sequences comprising more than one selected gene. In accordance with the disclosure, this could be used to introduce genetic material corresponding to an entire biosynthetic pathway into a plant. Introduction of such sequences may be facilitated by use of bacterial or yeast artificial chromosomes (BACs or YACs, respectively), or even plant artificial chromosomes. For example, the use of BACs for *Agrobacterium*-mediated transformation was disclosed by Hamilton et al. (1996).

[0139] Particularly useful for transformation are expression cassettes that have been isolated from such vectors. DNA segments used for transforming plant cells will, of course, generally comprise the cDNA, gene or genes which one desires to introduce into and have expressed in the host cells. These DNA segments can further include structures such as promoters, enhancers, polylinkers, or even regulatory genes as desired. The DNA segment or gene chosen for cellular introduction will often encode a protein which will be expressed in the resultant genetically modified cells resulting in a screenable or selectable trait and/or will impart an improved phenotype to the resulting genetically modified plant. However, this may not always be the case, and the present disclosure also encompasses genetically modified plants incorporating non-expressed transgenes.

[0140] In accordance with the disclosure, a nucleic acid vector comprising a coding sequence may be introduced into a plant such as a *Citrus* tree or variety, such that, when the vector is transformed into a *Citrus* variety or plant as described herein, the coding sequence is expressed in the plant. In some embodiments the coding sequence may be expressed in, for example, the phloem or roots of the plant, or any other part of the plant. Expression of the coding sequence in the resulting genetically modified *Citrus* tree or variety results in the tree exhibiting increased tolerance or resistance to HLB when compared to a tree lacking expression of the coding sequence.

A. Proteins and Recombinant DNA Molecules

[0141] As used herein, a “protein/Coding DNA molecule” or “polypeptide/Coding DNA molecule” refers to a DNA molecule comprising a nucleotide sequence that encodes a protein or polypeptide. A “coding sequence” or “protein/Coding sequence” or “polypeptide/Coding sequence” means a DNA sequence that encodes a protein or polypeptide. A “sequence” means a sequential arrangement of nucleotides or amino acids. The boundaries of a protein/Coding sequence or polypeptide/Coding sequence are usually determined by a translation start codon at the 5′-terminus and a translation stop codon at the 3′-terminus. A protein/Coding molecule or polypeptide/Coding molecule may comprise a DNA sequence encoding a protein or polypeptide sequence. As used herein, “transgene expression,” “expressing a transgene,” “protein expression,” “polypeptide expression,” “expressing a protein,” and “expressing a polypeptide” mean the production of a protein or polypeptide through the process of transcribing a DNA molecule into messenger RNA (mRNA) and translating the mRNA into polypeptide chains, which may be ultimately folded into proteins. A protein/Coding DNA molecule or polypeptide/Coding DNA molecule may be operably linked to a heterologous promoter in a DNA construct for use in expressing the protein or polypeptide in a cell transformed with the recombinant DNA molecule. As used herein, “operably linked” means two DNA molecules linked in manner so that one may affect the function of the other. Operably-linked DNA molecules may be part of a single contiguous molecule and may or may not be adjacent. For example, a promoter is operably linked with a protein/Coding DNA molecule or polypeptide/Coding DNA molecule in a DNA construct where the two DNA molecules are so arranged that the promoter may affect the expression of the transgene.

[0142] As used herein, a “DNA construct” is a recombinant DNA molecule comprising two or more heterologous DNA sequences. DNA constructs are useful for transgene expression and may be comprised in vectors and plasmids. DNA constructs may be used in vectors for the purpose of genome modification, that is the introduction of heterologous DNA into a host cell, in order to produce genetically modified plants and cells, and as such may also be contained in the plastid DNA or genomic DNA of a genetically modified plant, seed, cell, or plant part. As used herein, a “vector” means any recombinant DNA molecule that may be used for the purpose of genetically modifying a plant or plant cell. Recombinant DNA molecules as set forth in the sequence listing, can, for example, be inserted into a vector as part of a construct having the recombinant DNA molecule operably linked to a promoter that functions in a plant to drive expression of the protein encoded by the recombinant DNA molecule. Methods for constructing DNA constructs and vectors are well known in the art. The components for a DNA construct, or a vector comprising a DNA construct, generally include, but are not limited to, one or more of the following: a suitable promoter for the expression of an operably linked DNA, an operably linked protein/Coding DNA molecule, and a 3′ untranslated region (3′-UTR). Promoters useful in practicing the present disclosure include those that function in a plant for expression of an operably linked polynucleotide. Such promoters are varied and well known in the art and include those that are inducible, viral, synthetic, constitutive, temporally regulated, spatially regulated, and/or spatio-temporally regulated. Additional optional components include, but are not limited to, one or more of the following elements: 5′-UTR, enhancer, leader, cis-acting element, intron, chloroplast transit peptides (CTP), and one or more selectable marker transgenes.

[0143] Recombinant DNA molecules of the present disclosure may be synthesized and modified by methods known in the art, either completely or in part, especially where it is desirable to provide sequences useful for DNA manipulation (such as restriction enzyme recognition sites or recombination-based cloning sites), plant-preferred sequences (such as plant/Codon usage or Kozak consensus sequences), or sequences useful for DNA construct design (such as spacer or linker sequences). The present disclosure includes recombinant DNA molecules and proteins having at least about 80% (percent) sequence identity, about 81% sequence identity, about 82% sequence identity, about 83% sequence identity, about 84% sequence identity, about 85% sequence identity, about 86% sequence identity, about 87% sequence identity, about 88% sequence identity, about 89% sequence identity, about 90% sequence identity, about 91% sequence identity, about 92% sequence identity, about 93% sequence identity, about 94% sequence identity, about 95% sequence identity, about 96% sequence identity, about 97% sequence identity, about 98% sequence identity, about 99% sequence identity, or about 100% sequence identity to a coding sequence provided herein, for instance the sequences set forth as SEQ ID NOs: 2-9-11-51. As used herein, the term "percent sequence identity" or "% sequence identity" refers to the percentage of identical nucleotides or amino acids in a linear polynucleotide or polypeptide sequence of a reference ("query") sequence (or its complementary strand) as compared to a test ("subject") sequence (or its complementary strand) when the two sequences are optimally aligned (with appropriate nucleotide or amino acid insertions, deletions, or gaps totaling less than 20 percent of the reference sequence over the window of comparison). Optimal alignment of sequences for aligning a comparison window are well known to those skilled in the art and may be conducted by tools such as the local homology algorithm of Smith and Waterman, the homology alignment algorithm of Needleman and Wunsch, the search for similarity method of Pearson and Lipman, and by computerized implementations of these algorithms such as GAP, BESTFIT, FASTA, and TFASTA available as part of the Sequence Analysis software package of the GCG® Wisconsin Package® (Accelrys Inc., San Diego, CA), MEGAlign (DNAStar, Inc., Madison, WI), and MUSCLE (version 3.6) (Edgar, *Nucl. Acids Res.* 32:1792-1797, 2004) with default parameters. An "identity fraction" for aligned segments of a test sequence and a reference sequence is the number of identical components which are shared by the two aligned sequences divided by the total number of components in the reference sequence segment, that is, the entire reference sequence or a smaller defined part of the reference sequence. Percent sequence identity is represented as the identity fraction multiplied by 100. The comparison of one or more sequences may be to a full-length sequence or a portion thereof, or to a longer sequence.

[0144] Proteins in accordance with the disclosure may be produced by changing (that is, modifying) a wild-type protein to produce a new protein with a novel combination of useful protein characteristics, such as altered Vmax, Km, substrate specificity, substrate selectivity, and protein stability. Modifications may be made at specific amino acid positions in a protein and may be a substitution of the amino acid found at that position in nature (that is, in the wild-type protein) with a different amino acid. Proteins provided by the disclosure thus provide a new protein with one or more altered protein characteristics relative to the wild-type protein found in nature. In one embodiment of the disclosure, a protein may have altered protein characteristics such as improved or decreased activity against one or more herbicides or improved protein stability as compared to a similar wild-type protein, or any combination of such characteristics. In one embodiment, the disclosure provides a protein, and the DNA molecule or coding sequence encoding it, having at least about 80% sequence identity, about 81% sequence identity, about 82% sequence identity, about 83% sequence identity, about 84% sequence identity, about 85% sequence identity, about 86% sequence identity, about 87% sequence identity, about 88% sequence identity, about 89% sequence identity, about 90% sequence identity, about 91% sequence identity, about 92% sequence identity, about 93% sequence identity, about 94% sequence identity, about 95% sequence identity, about 96% sequence identity, about 97% sequence identity, about 98% sequence identity, about 99% sequence identity, or about 100% sequence identity to a protein sequence such as set forth as SEQ ID NOs: 2-9 and SEQ ID NO:11-51. Amino acid mutations may be made as a single amino acid substitution in the protein or in combination with one or more other mutation(s), such as one or more other amino acid substitution(s), deletions, or additions. Mutations may be made as described herein or by any other method known to those of skill in the art

B. Regulatory Elements

[0145] The plants and methods of the present disclosure can utilize a vector comprising a coding sequence that, when the vector is transfected into a plant, the coding sequence is expressed in the plant. The site and conditions under which the first selected DNA is expressed can be controlled to a great extent by selecting a promoter element in the vector that causes expression under the desired conditions.

[0146] In some embodiments, the coding sequence is expressed primarily in the roots of the plant, or in the phloem tissue of the plant. In this case, the coding sequence may be expressed in a greater quantity in roots or phloem than in other tissues of the plant. In some embodiments, more than one copy of an coding sequence may be expressed in a plant such that expression in the roots or phloem may be at least twice as much as in any other individual plant tissue (e.g., leaves, flowers, etc).

[0147] Limiting expression of the coding sequence primarily to the roots or phloem of a plant may be accomplished by operably linking the coding sequence to a heterologous promoter active in plant tissues, such as a root-specific or phloem-specific promoter. In other embodiments, a constitutive promoter may be preferred such that the coding sequence is expressed in all tissues of the plant. In some embodiments, a phloem-specific promoter in accordance with the disclosure may comprise an *Arabidopsis* sucrose-proton symporter 2 (AtSUC2) promoter, or a constitutive promoter may comprise a CaMV 35S promoter. Any root-specific or phloem-specific promoter known in the art may potentially be utilized to direct expression of the coding sequence to the roots or the phloem tissue. Examples of these may include, but are not limited to, an RB7, RPE15, RPE14, RPE19, RPE29, RPE60, RPE2, RPE39, RPE61, SHR, ELG3, EXP7, EXP18 or Atlg73160 promoter (Vijaybhaskar et al., 2008; Kurata et al., 2005; PCT Publication WO 01/53502; U.S. Pat. No. 5,459,252; Cho and Cosgrove, 2002).

[0148] In some embodiments, a coding sequence as described herein may be expressed at any level in the plant such that it may be detected in the plant using techniques known in the art. A coding sequence may be expressed in a greater quantity in a genetically modified *Citrus* plant or variety than in a plant not expressing the coding sequence as described herein. In some embodiments, the coding sequence is expressed at least twice as much as in a plant not expressing a coding sequence. In further embodiments, the coding sequence is expressed at least three, or four, or five times, or more, as much as in a plant not expressing a coding sequence. In yet another embodiment, there is no detectable expression of the coding sequence in a plant not expressing a coding sequence.

[0149] The DNA sequence between the transcription initiation site and the start of the coding sequence, i.e., the untranslated leader sequence, can also influence gene expression. One may thus wish to employ a particular leader sequence with a transformation construct of the disclosure. Useful leader sequences are contemplated to include those which comprise sequences predicted to direct optimum expression of the attached gene, i.e., to include a consensus leader sequence which may increase or maintain mRNA stability and prevent inappropriate initiation of translation. The choice of such sequences will be known to those of skill in the art in light of the present disclosure.

[0150] It is contemplated that vectors for use in accordance with the present disclosure may be constructed to include an ocs enhancer element. This element was first identified as a 16-bp palindromic enhancer from the octopine synthase (ocs) gene of *Agrobacterium* (Ellis et al., 1987), and is present in at least 10 other promoters (Bouchez et al., 1989). The use of an enhancer element, such as the ocs element and particularly multiple copies of the element, may act to increase the level of transcription from adjacent promoters when applied in the context of plant transformation.

C. Terminators

[0151] Transformation constructs prepared in accordance with the disclosure will typically include a 3′ end DNA sequence that acts as a signal to terminate transcription and allow for the poly-adenylation of the mRNA produced by coding sequences operably linked to a promoter. In one embodiment of the disclosure, the native terminator of a coding sequence coding sequence may be used. Alternatively, a heterologous 3′ end may enhance the expression of coding sequences. Examples of terminators that are deemed to be useful in this context include those from the nopaline synthase gene of *Agrobacterium tumefaciens* (nos 3′ end) (Bevan et al., 1983), the terminator for the T7 transcript from the octopine synthase gene of *Agrobacterium tumefaciens,* and the 3′ end of the protease inhibitor I or II genes from potato or tomato. Regulatory elements such as an Adh intron (Callis et al., 1987), sucrose synthase intron (Vasil et al., 1989) or TMV omega element (Gallie et al., 1989), may further be included where desired.

D. Transit or Signal Peptides

[0152] Sequences that are joined to the coding sequence of an expressed gene, which are removed post-translationally from the initial translation product and which facilitate the transport of the protein into or through intracellular or extracellular membranes, are termed transit (usually into vacuoles, vesicles, plastids and other intracellular organelles) and signal sequences (usually to the endoplasmic reticulum, Golgi apparatus, and outside of the cellular membrane). By facilitating the transport of the protein into compartments inside and outside the cell, these sequences may increase the accumulation of gene product protecting them from proteolytic degradation. These sequences also allow for additional mRNA sequences from highly expressed genes to be attached to the coding sequence of the genes. Since mRNA being translated by ribosomes is more stable than naked mRNA, the presence of translatable mRNA in front of the gene may increase the overall stability of the mRNA transcript from the gene and thereby increase synthesis of the gene product. Since transit and signal sequences are usually post-translationally removed from the initial translation product, the use of these sequences allows for the addition of extra translated sequences that may not appear on the final polypeptide. It further is contemplated that targeting of certain proteins may be desirable in order to enhance the stability of the protein (U.S. Pat. No. 5,545,818, incorporated herein by reference in its entirety).

[0153] Additionally, vectors may be constructed and employed in the intracellular targeting of a specific gene product within the cells of a genetically modified plant or in directing a protein to the extracellular environment. This generally will be achieved by joining a DNA sequence encoding a transit or signal peptide sequence to the coding sequence of a particular gene. The resultant transit, or signal, peptide will transport the protein to a particular intracellular, or extracellular destination, respectively, and will then be post-translationally removed.

E. Marker Genes

[0154] By employing a selectable or screenable marker protein, one can provide or enhance the ability to identify transformants. “Marker genes” are genes that impart a distinct phenotype to cells expressing the marker protein and thus allow such transformed cells to be distinguished from cells that do not have the marker. Such genes may encode either a selectable or screenable marker, depending on whether the marker confers a trait which one can “select” for by chemical means, i.e., through the use of a selective agent (e.g., a herbicide, antibiotic, or the like), or whether it is simply a trait that one can identify through observation or testing, i.e., by “screening” (e.g., the green fluorescent protein). Many examples of suitable marker proteins are known to the art and can be employed in the practice of the disclosure. Examples include, but not limited to, neo (Potrykus et al., 1985), bar (Hinchee et al., 1988), bxn (Stalker et al., 1988); a mutant acetolactate synthase (ALS) (European Patent Application 154, 204, 1985) a methotrexate resistant DHFR (Thillet et al., 1988), β-glucuronidase (GUS); R-locus (Dellaporta et al., 1988), β-lactamase (Sutcliffe, 1978), xy/E (Zukowsky et al., 1983), α-amylase (Ikuta et al., 1990), tyrosinase (Katz et al., 1983), β-galactosidase, luciferase (lux) (Ow et al., 1986), acquorin (Prasher et al., 1985), and green fluorescent protein (Sheen et al., 1995; Haseloff et al., 1997; Reichel et al., 1996; Tian et al., 1997; WO 97/41228).

[0155] Included within the terms “selectable” or “screenable” markers also are genes which encode a “secretable marker” whose secretion can be detected as a means of identifying or selecting for genetically modified cells. Examples include markers which are secretable antigens that can be identified by antibody interaction, or even secretable enzymes which can be detected by their catalytic activity. Secretable proteins fall into a number of classes, including small, diffusible proteins detectable, e.g., by ELISA; small active enzymes detectable in extracellular solution (e.g., α-amylase, β-lactamase, phosphinothricin acetyltransferase); and proteins that are inserted or trapped in the cell wall (e.g., proteins that include a leader sequence such as that found in the expression unit of extensin or tobacco PR-S).

IV. Antisense and RNAi Constructs

[0156] In the methods and compositions of the present disclosure, endogenous gene activity can be down-regulated by any means known in the art, including through the use of ribozymes or aptamers. Endogenous gene activity can also be down-regulated with an antisense or RNAi molecule.

[0157] In particular, constructs comprising a coding sequence, including fragments thereof, in antisense orientation, or combinations of sense and antisense orientation, may be used to decrease or effectively eliminate the expression of the gene in a plant such as a *Citrus* tree or variety. Accordingly, this may be used to “knock-out” the function of the coding sequence or homologous sequences thereof.

[0158] Techniques for RNAi are well known in the art and are described in, for example, Lehner et al., (2004) and Downward (2004). The technique is based on the ability of double stranded RNA to direct the degradation of messenger RNA with sequence complementary to one or the other strand (Fire et al., 1998). Therefore, by expression of a particular coding sequence in sense and antisense orientation, either as a fragment or longer portion of the corresponding coding sequence, the expression of that coding sequence can be down-regulated.

[0159] Antisense, and in some aspects RNAi, methodology takes advantage of the fact that nucleic acids tend to pair with “complementary” sequences. By complementary, it is meant that polynucleotides are those which are capable of base-pairing according to the standard Watson/Crick complementarity rules. That is, the larger purines will base pair with the smaller pyrimidines to form combinations of guanine paired with cytosine (G:C) and adenine paired with either thymine (A:T) in the case of DNA, or adenine paired with uracil (A:U) in the case of RNA. Inclusion of less common bases such as inosine, 5-methylcytosine, 6-methyladenine, hypoxanthine and others in hybridizing sequences does not interfere with pairing.

[0160] Targeting double-stranded (ds) DNA with polynucleotides leads to triple-helix formation; targeting RNA will lead to double-helix formation. Antisense oligonucleotides, when introduced into a target cell, specifically bind to their target polynucleotide and interfere with transcription, RNA processing, transport, translation and/or stability. Antisense and RNAi constructs, or DNA encoding such RNA's, may be employed to inhibit gene transcription or translation or both within a host cell, either in vitro or in vivo, such as within a host plant cell. In certain embodiments of the disclosure, such an oligonucleotide may comprise any unique portion of a nucleic acid sequence provided herein. In certain embodiments of the disclosure, such a sequence comprises at least 18, 30, 50, 75, or 100 or more contiguous nucleic acids of the nucleic acid sequence of a gene, and/or complements thereof, which may be in sense and/or antisense orientation. By including sequences in both sense and antisense orientation, increased suppression of the corresponding coding sequence may be achieved.

[0161] Constructs may be designed that are complementary to all or part of the promoter and other control regions, exons, introns or even exon-intron boundaries of a gene. It is contemplated that the most effective constructs may include regions complementary to intron/exon splice junctions. Thus, it is proposed that an embodiment includes a construct with complementarity to regions within 50-200 bases of an intron-exon splice junction. It has been observed that some exon sequences can be included in the construct without seriously affecting the target selectivity thereof. The amount of exonic material included will vary depending on the particular exon and intron sequences used. One can readily test whether too much exon DNA is included simply by testing the constructs in vitro to determine whether normal cellular function is affected or whether the expression of related genes having complementary sequences is affected.

[0162] As stated above, “complementary” or “antisense” means polynucleotide sequences that are substantially complementary over their entire length and have very few base mismatches. For example, sequences of fifteen bases in length may be termed complementary when they have complementary nucleotides at thirteen or fourteen positions. Naturally, sequences which are completely complementary will be sequences which are entirely complementary throughout their entire length and have no base mismatches. Other sequences with lower degrees of homology also are contemplated. For example, an RNAi or antisense construct which has limited regions of high homology, but also contains a non-homologous region (e.g., as in a ribozyme) could be designed. Methods for selection and design of sequences that generate RNAi are well known in the art (e.g. Reynolds, 2004). These molecules, though having less than 50% homology, would bind to target sequences under appropriate conditions.

[0163] It may be advantageous to combine portions of genomic DNA with cDNA or synthetic sequences to generate specific constructs. For example, where an intron is desired in the ultimate construct, a genomic clone will need to be used. The cDNA or a synthesized polynucleotide may provide more convenient restriction sites for the remaining portion of the construct and, therefore, would be used for the rest of the sequence. Constructs useful for generating RNAi may also comprise concatemers of sub-sequences that display gene regulating activity.

V. Methods for Genetic Transformation

[0164] Suitable methods for transformation of plant or other cells for use with the current disclosure are believed to include virtually any method by which DNA can be introduced into a cell, such as by direct delivery of DNA such as by PEG-mediated transformation of protoplasts (Omirullch et al., 1993), by desiccation/inhibition-mediated DNA uptake (Potrykus et al., 1985), by electroporation (U.S. Pat. No. 5,384,253, specifically incorporated herein by reference in its entirety), by *Agrobacterium*-mediated transformation (U.S. Pat. Nos. 5,591,616 and 5,563,055; both specifically incorporated herein by reference) and by acceleration of DNA coated particles (U.S. Pat. Nos. 5,550,318; 5,538,877; and 5,538,880; each specifically incorporated herein by reference in its entirety), etc. Through the application of techniques such as these, the cells of virtually any plant species may be stably transformed, and these cells developed into genetically modified plants.

[0165] *Agrobacterium*-mediated transfer is a widely applicable system for introducing genes into plant cells because the DNA can be introduced into whole plant tissues, thereby bypassing the need for regeneration of an intact plant from a protoplast. The use of *Agrobacterium*-mediated plant integrating vectors to introduce DNA into plant cells is well known in the art. See, for example, the methods described by Fraley et al., (1985), Rogers et al., (1987) and U.S. Pat. No. 5,563,055, specifically incorporated herein by reference in its entirety.

[0166] Another method for delivering transforming DNA segments to plant cells in accordance with the disclosure is microprojectile bombardment (U.S. Pat. Nos. 5,550,318; 5,538,880; 5,610,042; and PCT Application WO 94/09699; each of which is specifically incorporated herein by reference in its entirety). In this method, particles may be coated with nucleic acids and delivered into cells by a propelling force.

VI. Production and Characterization of Genetically Modified Plants

[0167] After effecting delivery of exogenous DNA to recipient cells, the next steps generally concern first identifying and selecting the transformed cells and from those cells identifying the selecting the genetically modified cells for further culturing and plant regeneration. In order to improve the ability to identify transformed and genetically modified cells, one may desire to employ one or more selectable or screenable marker genes with a transformation vector prepared in accordance with the disclosure. In this case, one would then generally assay the potentially transformed and modified cell population by exposing the cells to a selective agent or agents, or one would screen the cells for the desired marker gene trait.

[0168] It is believed that DNA is introduced into only a small percentage of target cells in any one study. In order to provide an efficient system for identification of those cells that are transformed and predisposed to genetic modification one may employ a means for selecting those cells that are stably transformed. One exemplary embodiment of such a method is to introduce, into the host cell, a marker gene which confers resistance to some normally inhibitory agent, such as an antibiotic or herbicide. Potentially transformed cells then are exposed to the selective agent. In the population of surviving cells will be those cells where, generally, the resistance/Conferring gene has been integrated and expressed at sufficient levels to permit cell survival. Cells may be tested further to confirm stable integration of the exogenous DNA.

[0169] Cells that survive the exposure to the selective agent, or cells that have been scored positive in a screening assay, may then be selected again using a second, distinct selection paradigm that detects those cells that contain the genetic modification. Cells that survive the exposure to the second selective agent, or cells that have been scored positive in the second screening assay, may be cultured in media that supports regeneration of plants. The genetically modified cells, identified by selection or screening and cultured in an appropriate medium that supports regeneration, will then be allowed to mature into plants. Developing plantlets are transferred to soilless plant growth mix, and hardened, e.g., in an environmentally controlled chamber, for example, at about 85% relative humidity, 600 ppm CO_2, and 25-250 microeinsteins $m^{-2}s^{-1}$ of light. Plants may be matured in a growth chamber or greenhouse. Plants can be regenerated from about 6 wk to 10 months after a genetically modified cell is identified, depending on the initial tissue.

[0170] To confirm the presence of the genetic modification in the regenerating plants, a variety of assays may be performed. Such assays include, for example, “molecular biological” assays, such as Southern and northern blotting and polymerase chain reaction (PCR); “biochemical” assays, such as detecting the absence or presence of a protein product, e.g., by immunological means (ELISAs and western blots) or by enzymatic function; plant part assays, such as leaf or root assays; and also, by analyzing the phenotype of the whole regenerated plant. Modification of the host genome and the independent identities of genetically modified plants may be determined using, e.g., Southern hybridization or PCR. Genetic modifications that affect, for example, protein or gene expression may then be evaluated by specifically measuring the expression of those affected molecules or evaluating the phenotypic changes brought about by their expression change.

VII. Breeding Plants of the Disclosure

[0171] In addition to direct transformation of a particular plant genotype with a construct prepared according to the current disclosure, genetically modified plants may be made by crossing a plant having a selected genetic modification of the disclosure to a second plant lacking the construct. For example, a selected lignin biosynthesis coding sequence can be introduced into a particular plant variety by crossing, without the need for ever directly transforming a plant of that given variety. Therefore, the current disclosure not only encompasses a plant directly modified or regenerated from cells which have been modified in accordance with the current disclosure, but also the progeny of such plants.

[0172] As used herein the term “progeny” denotes the offspring of any generation of a parent plant prepared in accordance with the instant disclosure, wherein the progeny comprises a selected DNA construct. “Crossing” a plant to provide a plant line having one or more added transgenes relative to a starting plant line, as disclosed herein, is defined as the techniques that result in a coding sequence of the disclosure being introduced into a plant line by crossing a starting line with a donor plant line that comprises a first selected DNA of the disclosure. To achieve this in a plant such as a *Citrus* tree one could, for example, perform the following steps: [0173] (a) plant seeds of the first (starting line) and second (donor plant line that comprises a first selected DNA of the disclosure) parent plants; [0174] (b) grow the seeds of the first and second parent plants into plants that bear flowers; [0175] (c) pollinate a flower from the first parent plant with pollen from the second parent plant; and [0176] (d) harvest seeds produced on the parent plant bearing the fertilized flower.

[0177] Backcrossing is herein defined as the process including the steps of: [0178] (a) crossing a plant of a first genotype containing a desired gene, DNA sequence or element to a plant of a second

genotype lacking the desired gene, DNA sequence or element; [0179] (b) selecting one or more progeny plant containing the desired gene, DNA sequence or element; [0180] (c) crossing the progeny plant to a plant of the second genotype; and [0181] (d) repeating steps (b) and (c) for the purpose of transferring a desired DNA sequence from a plant of a first genotype to a plant of a second genotype.
[0182] Introgression of a DNA element into a plant genotype is defined as the result of the process of backcross conversion. A plant genotype into which a DNA sequence has been introgressed may be referred to as a backcross converted genotype, line, inbred, or hybrid. Similarly a plant genotype lacking the desired DNA sequence may be referred to as an unconverted genotype, line, inbred, or hybrid.
[0183] In some embodiments, asexual reproduction or propagation may be used to obtain a progeny plant in accordance with the disclosure. Techniques to achieve asexual propagation or reproduction in *Citrus* trees or varieties may include, for example, grafting, budding, top-working, layering, runner division, cuttings, rooting, T-budding, and the like. In some embodiments, one *Citrus* variety into which a coding sequence has been introduced may be grafted onto the rootstock of another variety. In other embodiments, a coding sequence may be introduced into the rootstock. In either of these situations, one or both of the plant varieties may exhibit increased tolerance or resistance to HLB.
EXAMPLES
Example 1
Identification of Las SDEs
[0184] Las has been predicted to produce at least 166 SDEs, 36 of which have been shown to be highly expressed in plants. These 36 Las SDEs were stably or transiently expressed in *Citrus* plants and *Nicotiana benthamiana* to determine if they are HLB effectors. Transgenic expression of CLIBASIA_04025 (Las4025), CLIBASIA_00470 (Las470), CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250) induced symptoms consistent with HLB. For example, transgenic expression of CLIBASIA_04025 (Las4025) without its signal peptide stunted leaf growth, delayed plant growth, and induced leaf yellowing (FIG. **1**A and FIG. **1**B). The CLIBASIA_ 04025 expression in transgenic lines was confirmed using an antibody against CLIBASIA_ 04025 (FIG. **1**C). Transgenic expression of CLIBASIA_00470 also delayed plant growth.
Example 2
Las SDE and Host Target Proteins Interaction Assays
[0185] Yeast two-hybrid (Y2H) screening was performed to identify putative target proteins of the SDEs identified above. A cDNA library was generated from mRNA isolated from ‘Valencia’ sweet orange plants infected with Las. The mRNA was obtained from these plants during the early stage of infection in which the Las Ct value was between 28-30. The cDNA library was constructed using the Make Your Own Mate & Plate™ Library System (Clontech) following the manufacturer's instructions and had a titer greater than 3×10.sup.8 cfu.
[0186] The coding sequences of CLIBASIA_04025 (Las4025), CLIBASIA_00470 (Las470), CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250), without their signal peptides, were cloned in-frame with the GAL4 DNA-binding domain (BD) of the bait vector pGBKT7. The Y2H screen was performed using the Matchmaker® Gold Yeast Two-Hybrid System (Clontech) following the manufacturer's instructions. The SDE target proteins identified in the screen are summarized in Table 1B.
TABLE-US-00002 TABLE 1B Y2H-confirmed Las SDE target proteins. SDEs Target proteins CLIBASIA_04025 PP2-B2/12 (orange1.1t04174) (Las4025) Lectin (orange1.1t05126) Cysteine protease (Cs4g07410) Cysteine protease 15A-like (Cs3g25530) Myb family transcription factor (orange1.1t02260) YLS9-like (Cs2g29120) Cell death suppressor protein Lls1(Cs9g02990) Red chlorophyll catabolite reductase; Accelerated cell death 2 (Cs1g22670) Homolog to Acd2, red

chlorophyll catabolite reductase-like (Cs1g22680), which is 64% similar Homolog to Lls1 - Acd1-Like Cs9g03000, which is 93.4% similar; Acd1 Cs8g15480, which is 51% similar, but is also work in same chlorophyll catabolism pathway to Acd2. Homologs to Cysteine protease (Cs3g25530) and Cysteine Protease (Cs4g07410), all Papain-like cysteine proteases Homolog YLS9-like, NDR1/HIN1-like protein 13, Cs8g01640, which is 75% similar Homologs to Myb family transcription factor, PHL5-like, Cs7g01290, 56% similar; PHL5, orange1.1t02259, which is 76% similar Galactinol-sucrose galactosyltransferase 2 (Cs9g12460) Vacuolar protein sorting-associated protein 36 (Cs7g24050) DnaJ protein homolog (Cs7g23510) Plastid-specific ribosomal protein 4 (Cs6g08000) Pathogenesis-related protein 10 (Cs9g03630) Glucan endo-1,3-beta-D-glucosidase-like protein (orange1.1t00643) Core-2/I-branching beta-1,6-N-acetylglucosaminyltransferase family protein (Cs7g07430) Leucyl-tRNA synthetase bacterial/mitochondrial, class Ia (Cs2g02720) Annexin D1 (Cs3g18360) Pentatricopeptide repeat-containing protein (Cs5g26120) Probable fructose-bisphosphate aldolase 2, chloroplastic (Cs8g08710) Arginine/serine-rich splicing factor, putative, expressed (Cs3g18350) SVP1-like protein 2 (Cs5g32770) Alanine aminotransferase 2 (Cs7g09270) BEL1-like homeodomain protein 1 (Cs6g13660) AT-rich interactive domain-containing protein 4 (Cs4g06750) Heat shock factor protein HSF8 (Cs7g24140) Plasma membrane ATPase 1 (Cs6g03480) Gag-pol polyprotein (Cs7g14770) Phospholipid: diacylglycerol acyltransferase (Cs1g17750) Aconitate hydratase, cytoplasmic (Cs2g21430) DNA-directed RNA polymerase subunit alpha (orange1.1t03665) Polyubiquitin 10 (Cs4g11190) Diacylglycerol kinase theta (Cs4g02800) Chloroplast methionine sulfoxide reductase B2 (Cs9g05400) Leucine-rich repeat receptor protein kinase EXS (Cs7g18050) Lateral organ boundaries-domain 29 (orange1.1t00246.1) Formamidase (Cs1g21820) Signal peptidase complex subunit 3B (Cs3g13460) Probable plastid-lipid-associated protein 8 (Cs7g07440) Stress responsive gene 6 protein (orange1.1t01091) UBX domain-containing protein (Cs5g01690) Protein SRG1 (Cs5g13180) Thioredoxin H-type (Cs1g24740) Glycoside hydrolase (Cs8g12020) Thioredoxin F2 (Cs6g02830) RNA pseudourine synthase 7 (orange1.1t02625) Kunitz-type protease inhibitor KPI-D2.2 (Cs5g16850) Aspartate aminotransferase (Cs4g19830) UDP-glucuronate decarboxylase 4 (Cs6g05450) Nitrate transporter 1.5 (orange1.1t00223) FKBP-like peptidyl-prolyl cis-trans isomerase family protein (orange1.1t00062) Zinc-binding alcohol dehydrogenase domain-containing protein 2 (Cs8g05790) Mannitol dehydrogenase (Cs1g20600) RNA recognition motif family protein (Cs2g07940) alpha/beta-Hydrolases superfamily protein (Cs2g21120) Protein argonaute 1 (Cs5g16710) Progesterone 5-beta-reductase (Cs3g11840) NADH dehydrogenase [ubiquinone] 1 alpha subcomplex subunit 6 (Cs1g16240) Tetratricopeptide repeat (TPR)-like superfamily protein (Cs6g03690) Mitochondrial carrier domain-containing protein (Cs6g03800) Cell division control protein 48 homolog C (Cs3g01650) Subtilisin-like protease (Cs8g02780) U6 snRNA-associated Sm-like protein LSm6 (orange1.1t02120) Subtilisin-like protease (Cs8g06090) Histone H4 (Cs8g18120) Chromatin-associated protein Dek (Cs4g08790) Serine carboxypeptidase-like 49 (Cs7g24460) RING/U-box superfamily protein (Cs8g16720) Linoleate 13S-lipoxygenase 2-1 (orange1.1t04376) Isoflavone reductase-like (Cs2g16260) alpha-like protein (Cs6g16290.1) THO complex subunit 3 (Cs7g18110) Uncharacterized protein Sb07g024435 (Cs7g26460) Ubiquitin-activating enzyme E1 2 (Cs8g20660) Structure-specific endonuclease subunit SLX1 (Cs2g09350) Putative uncharacterized protein P0458H05.117 (Cs4g05300) Aconitate hydratase 1 (Cs1g26040) DnaJ homolog subfamily B member 13 (Cs3g04780) Plastocyanin (Cs3g26730) Putative uncharacterized protein Sb03g000880 (Cs5g31880) 50S ribosomal protein L14 (orange1.1t04817) Oligopeptidase A (Cs1g20720) Maturase K (Cs2g09070) Protein FRA10AC1 (Cs7g01730) Putative Uncharacterized protein AlNc14C124G6763 (Cs8g09030) Putative Uncharacterized protein Sb10g020525 (orange1.1t00482) Putative Uncharacterized protein OSJNBb0021O11.27 (Cs4g16820) CLIBASIA_00470 Lectin (orange1.1t05126) (Las470) Galactinol-sucrose galactosyltransferase 2-like (Cs9g12460) DnaJ homolog 1 like (Cs1g19720) YLS9-like (Cs2g29120) 8-hydroxygeraniol dehydrogenase (XM_006466284.2) CLIBASIA_04065

Hypothetical protein (orange1.1t00563) (Las4065) CLIBASIA_05150 Cysteine protease (Cs4g07410) (Las5150) CLIBASIA_04250 Translation initiation factor IF-3 (orange1.1g044576m) (Las4250)

[0187] Multiple proteins interacted with CLIBASIA_04025 (Las4025) and CLIBASIA_00470 (Las470); whereas, CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250) each interacted with a single protein. CLIBASIA_04025 (Las4025) interacted with PP2-B2 (orange1.1t04174), a phloem protein. Phloem protein encoding genes are known to be involved in phloem blockage and are suggested to contribute to HLB symptom onset. In addition, SDE15 was shown to interact with the CtACD2 protein by the Y2H assay (FIG. **5**A).

[0188] To confirm the initial Y2H screen results, full-length sequences of the SDE target proteins were cloned in-frame with the GALA activation domain (AD) of the prey vector pGADT7. Following the manufacturer's instructions, the Y2HGOLD yeast strain was co-transformed with relevant bait and prey vector pairs. For negative controls, a prey vector co-transformed with an empty bait vector was used. Exemplary results that were achieved in the Y2H assay using the CLIBASIA_04025 (Las4025) bait vector are shown in FIG. **2**. CLIBASIA_04025 (Las4025) interacted with PP2-B2 (orange1.1t04174), and Pathogenesis-related protein 10 (Cs9g03630).

[0189] Glutathione S-transferase (GST) pull-down and Bimolecular fluorescence complementation (BiFC) assays were performed to further confirm Y2H results. Las genomic DNA was extracted from infected *Citrus* leaves and the coding sequences of CLIBASIA_04025 (Las4025), CLIBASIA_00470 (Las470), CLIBASIA_04065 (Las4065), CLIBASIA_05150 (Las5150), and CLIBASIA_04250 (Las4250) were PCR-amplified for use in both assays. For the GST pull-down assay, the respective fragments were cloned in-frame with Maltose-binding protein (MBP) in the pMAL™/C5X vector (NEB, USA) to generate MBP-SDE fusion proteins. The coding sequences of the SDE target proteins were PCR-amplified using *Citrus* leaf cDNA as a template. The respective fragments were cloned in-frame with GST in the pGEX-4T-1 vector (GE Healthcare, USA) to generate GST-target fusion proteins. For the BiFC assays, the coding sequences of the SDE target proteins were PCR-amplified using *Citrus* leaf cDNA as a template. The respective SDEs and SDE target proteins were cloned in-frame with either N-terminal or C-terminal fragments of EYFP using pSAT6-nEYFP/C1 and pSAT6/CEYFP/C1-B vectors, respectively. This was done with an In-Fusion cloning kit (Clontech, USA) and produced SDE-EYFPN, EYFPC-SDE, SDE target-EYFPN, and EYFPC-SDE target fusion proteins. Citrus protoplasts isolated from grapefruit epicotyl segments were co-transformed with pairs of EYFPC and EYFPN vectors. Additionally, SDE15 was shown to interact with the CtACD2 protein by the GST pull-down assay (FIG. **5**C) and the BiFC assay (FIG. **5**B).

[0190] Exemplary results obtained in the GST pull-down and BiFC assays are provided in FIG. **3**. In both assays, the CLIBASIA_04025 (Las4025) directly interacted with cysteine protease (Cs4g07410), therefore confirming the interaction observed between the two proteins in the Y2H assays. No interactions were detected in any of the negative controls (FIG. **3**A and FIG. **3**B).

[0191] SDE15 was also shown to interact with the CtACD2 protein that negatively regulates the hypersensitive reaction by the hypersensitive response (HR) assay (FIG. **5**D). The electrolyte leakage associated with HR induced by AvrBsT protein is shown in FIG. **5**E.

Example 3

Overexpression of SDE Target Proteins in Citrus Plants

[0192] Overexpression of Myb family transcription factor (orange1.1t02260), a CLIBASIA_04025 (Las4025) target protein, in Duncan grapefruits induced symptoms similar to those observed in *Citrus* plants infected with *Candidatus Liberibacter asiaticus*. Plants that overexpressed MYTL displayed stunted plant growth and greater starch accumulation when compared to wild type controls (FIG. **4**A, FIG. **4**B, and FIG. **4**C).

Example 4

Modification of Susceptibility Genes in Citrus Plants

[0193] Based on the identification of susceptibility genes that encode proteins that are targets of SDEs secreted by Las, *Citrus* plants and regulatory elements thereof may be modified to exhibit resistance to infection by Las. In particular, plants may be modified to exhibit reduced expression of one or more of the susceptibility genes identified herein. These susceptibility genes and regulatory elements thereof may themselves be modified using targeted genome editing (e.g., single-stranded oligonucleotide-mediated genome editing, meganucleases, zinc finger nucleases, TALENs, or RNA-guided nuclease (e.g., CRISPR technology). Modified *Citrus* plants may be produced using a plant transformation vector comprising an expression cassette comprising a promoter and 3′ UTR operably linked to the coding sequence for a modified target protein. The plant transformation vector may be used with *Agrobacterium tumefaciens* and standard methods for plant transformation known in the art.

[0194] Alternatively, plants comprising modified endogenous genes or regulatory elements thereof may be produced by inserting DNA directly into the plant genome at a specified targeted location. Any site or locus within the plant genome may potentially be chosen for site-directed integration of a transgene or construct of the present disclosure. For site-directed integration, a double-strand break (DSB) or nick may first be made at a selected genomic locus with a site-specific nuclease, such as, for example, a zinc-finger nuclease, an engineered or native meganuclease, a TALE-endonuclease, or an RNA-guided endonuclease (e.g., Cas9, Cpf1, CasX or CasY). Any method known in the art for site-directed integration may be used. In the presence of a donor template molecule, the DSB or nick may then be repaired by homologous recombination between the homology arm(s) of the donor template and the plant genome, or by non-homologous end joining (NHEJ), resulting in site-directed integration of the insertion sequence into the plant genome to create the targeted insertion event at the site of the DSB or nick.

[0195] Plants having one or more mutations in the genomic susceptibility genes or regulatory elements thereof may be produced using a double-strand break (DSB) or nick made at the genomic locus with a site-specific nuclease, such as, a zinc-finger nuclease, an engineered or native meganuclease, a TALE-endonuclease, or an RNA-guided endonuclease (e.g., Cas9, Cpf1, CasX or CasY). Any method known in the art for genome editing or non-templated editing may be used. Delivery methods for the nuclease and gRNA include, but are not limited to, delivering by *Agrobacterium*-mediated methods, delivering as a protein or RNA using transfection or biolistics, and delivery by expression from a virus. One or more nucleases or gRNA may be used. Donor molecules to deliver the desired changes may include, but are not limited to, double-stranded DNA, single-stranded DNA oligonucleotides, RNA or viral DNA. Donor molecules may be delivered by *Agrobacterium*, virus, biolistic delivery, or transfection. In the presence of a donor template molecule, the one or more DSBs or nicks may then be repaired by homologous recombination between the homology arm(s) of the donor template and the plant genome, by non-homologous end joining (NHEJ), by single-strand annealing pathway or other DNA repair mechanisms resulting in modification of the native sequence in the plant genome to that contained by the donor to create the desired mutation.

[0196] Modified *Citrus* plants may be tested for resistance to HLB. Citrus plants that are modified to exhibit reduced expression of one or more of the susceptibility genes identified herein may be inoculated with a strain from the genus *Ca. Liberibacter* and evaluated for symptoms of HLB infection compared to *Citrus* plants that do not comprise modified susceptibility genes or regulatory elements thereof. Modified plants that test negative for HLB may be used for further greenhouse and field testing and breeding of HLB resistant *Citrus* lines.

Example 5

Overexpression of SDE15 in *Nicotiana tabacum* and *Citrus*

[0197] SDE15 (CLIBASIA_04025, 11.17 kDa) is a hypothetical protein. Sequence analysis of SDE15 did not reveal any insights into its function except the presence of a typical signal peptide (amino acids 1-22) predicted by SignalP v4.1 (FIG. **6**A). Interestingly, transgenic expression of

SDE15 without the signal peptide (8.68 kDa) in *N. tabacum* caused markedly reduced plant growth and a slightly yellow leaf phenotype, suggestive of biological effects of SDE15 on host biology. These interesting phenotypes, observed in a heterologous plant (*N. tabacum*), prompted us to overexpress SDE15 in the native host plant *Citrus* (cv. Duncan grapefruit) via stable *Agrobacterium* mediated transformation to determine whether SDE15 is involved in inducing HLB-like symptoms, suppressing plant defenses and/or promoting Las growth. Indeed, SDE15-transgenic Duncan grapefruit plants showed some HLB-like symptoms including slight yellowing, smaller leaves, and reduced growth at the early stage of development (FIG. **6**B). Interestingly, these HLB-like symptoms disappeared approximately one year after transgenic plants were grown under greenhouse conditions.

[0198] Las-infected plants show characteristic gene expression patterns (Albrecht, U., and Bowman, K. (2008). PLANT SCIENCE 175, 291-306; Kim, J. S., Sagaram, U. S., Burns, J. K., Li, J. L., and Wang, N. (2009). Phytopathology 99, 50-57; Albrecht, U., and Bowman, K. D. (2012). Plant Sci 185-186, 118-130; Rawat, N., Kiran, S. P., Du, D., Gmitter, F. G., and Deng, Z. (2015). BMC Plant Biol 15, 184). To determine whether SDE15 also modulates host gene expression, we compared the expression patterns of HLB-associated marker genes in SDE15-transgenic and EV (empty vector)-transgenic *Citrus* plants. We found that expression of the genes encoding phloem protein 2 (PP2)-B15, WRKY40, and Zn transporter 5 (ZIP5) was highly induced in SDE15-transgenic *Citrus*, whereas expression of the genes encoding sugar transporter protein Sweet7 and Sweet15 and chalcone synthase were reduced compared to control, resembling those in HLB plants for these marker genes (Albrecht and Bowman, 2008; Kim et al., 2009; Albrecht and Bowman, 2012; Rawat et al., 2015). Taken together, both the partial HLB-like symptom, albeit transient, and the expression pattern of HLB marker genes in the SDE15-transgenic *Citrus* indicate that SDE15 is involved in the outcome of the Las-*Citrus* interaction.

Example 6

Detection of SDE15 Inside Phloem Sap and Subcellular Localization of SDE15

[0199] To function as an effector, we expect that SDE15 would be secreted into the cytosol of phloem cells and may be detected in the phloem sap of Las-infected plants. We therefore developed a SDE15-specific antibody and examined whether SDE15 can be detected in the phloem sap by western blot. We could clearly detect SDE15 in the phloem sap extract of Las-infected barks but not in healthy *Citrus* barks by western blot (FIG. **6**C). To determine the subcellular locations of SDE15 in the plant cell, we performed subcellular localization of SDE15 by transient expression of SDE15-EYFP fusion protein in *N. benthamiana*. We found that SDE15 is localized to the cytoplasm and the nucleus (FIG. **6**D).

[0200] We next analyzed the expression pattern of the SDE15 transcript in Las-infected *Citrus* and psyllids using quantitative reverse transcription-PCR (qRT-PCR). SDE15 showed a higher expression level in *Citrus* than in psyllids (FIG. **6**E). Furthermore, SDE15 expression level was higher in the early stage (asymptomatic) than the later stage (symptomatic) of Las infection (FIG. **6**F), indicating that SDE15 likely functions at a relatively early stage of Las infection before the appearance of visible HLB symptoms.

Example 7

SDE15 Suppresses PCD and Promotes Las Growth in Planta

[0201] Many effectors of extracellular pathogens have been reported to suppress plant defenses (Jones, J. D., and Dangl, J. L. (2006). Nature 444, 323-329.; Toruño, T. Y., Stergiopoulos, I., and Coaker, G. (2016). Annu Rev Phytopathol 54, 419-441). To investigate whether SDE15 (from a cytosolic bacterial pathogen) is involved in suppressing plant defenses, we inoculated SDE15-transgenic and nontransgenic Duncan grapefruit with *Xanthomonas citri* subsp. *citri* strain A.sup.w (XccA.sup.w), which triggers the immunity-associated hypersensitive reaction (HR, a form of PCD) in Duncan grapefruit. Intriguingly, we found XccA.sup.w did not induce an HR in the leaves of SDE15-transgenic Duncan grapefruit, whereas an HR was observed in the nontransgenic

Duncan grapefruit (FIG. **7**A). We next examined immunity-associated gene expression in SDE15 transgenic and nontransgenic plants after inoculation with XccA.sup.w. The expression of PR1, PR3, and PR5 genes was markedly reduced in the SDE15-transgenic than nontransgenic Duncan grapefruit (FIG. **7**B). Taken together, these results suggest that SDE15 is a potent suppressor of PCD in *Citrus.*

[0202] The ability of SDE15 to suppress PCD and defense gene expression prompted us to examine whether the SDE15 transgenic plants could promote Las growth. For this purpose, we graft-inoculated SDE15-transgenic and EV-transgenic Duncan grapefruit plants with Las. Significant increase of Las titers was observed over a four-month period in the SDE15-transgenic *Citrus* compared to the EV-transgenic Duncan grapefruit plants, as estimated by qPCR assays (FIG. **8**). In accordance, visible HLB symptoms were observed earlier on leaves of SDE15-transgenic *Citrus* than control plants after Las graft-inoculation (FIG. **8**).

[0203] Because SDE15 suppressed PCD triggered by XccA.sup.w, we also tested whether SDE15 transgenic plants become more susceptible to virulent Xcc strain XacA 306. Indeed, the canker symptom caused by XacA 306 appeared faster in SDE15 transgenic *Citrus* compared to EV-transgenic plants. In addition, higher XacA 306 populations were detected in SDE15 transgenic grapefruit than that in the EV-transgenic Duncan grapefruit (FIG. **9**). Thus, transgenic expression of SDE5 causes broad susceptibility to both HLB and *Citrus* canker, two of the most important diseases in *Citrus.*

Example 8

SDE15 Interacts With the PCD Repressing Protein CsACD2

[0204] Because SDE15 does not contain obvious sequence domains that suggest enzymatic activities, we hypothesized that SDE15 might exert its suppressive effect on PCD and defense gene expression via protein-protein interactions. To identify the targets of SDE15, we performed yeast two-hybrid (Y2H) screening with SDE15 without its signal peptide using cDNA libraries generated with mRNAs isolated from the Valencia sweet orange at different Las infection stages (healthy (H), asymptomatic (AS) and symptomatic(S)). We identified 20 SDE15-interacting proteins in the H library, 60 in the AS library and 6 in the S library (Table 1B).

[0205] Among the SDE15-interacting proteins identified by Y2H (Table 1B), CsACD2 (Accession No. Cs1g22670) (FIG. **10**) was of special interest. It shares 55% similarity with the *Arabidopsis* ACD2 protein (Greenberg et al., 1994; Wüthrich et al., 2000). In *Arabidopsis,* ACD2protects cells from PCD caused by endogenous porphyrin-related molecules including red chlorophyll catabolite (RCC) or exogenous protoporphyrin IX (Mach, J. M., Castillo, A. R., Hoogstraten, R., and Greenberg, J. T. (2001). T Proc Natl Acad Sci U S A 98, 771-776; Yao, N., Eisfelder, B. J., Marvin, J., and Greenberg, J. T. (2004). Plant J 40, 596-610). We hypothesized that SDE15 may target CsACD2 to suppress PCD as a mechanism to promote Las growth in the phloem. We confirmed the SDE15-CsACD2 interaction via pair-wise Y2H assay (FIG. **10**A). To further confirm the interaction between SDE15 and CsACD2, we conducted bimolecular fluorescence complementation (BiFC) assay in vivo and glutathione-S-transferase (GST) pull-down assay in vitro. Co-transformation of SDE15-EYFPN and EYFPC-CsACD2 or EYFPC-SDE15 and CsACD2-EYFP.sup.N into *Citrus* leaf protoplasts displayed strong signals in the cytoplasm, but not in negative controls (FIG. **10**B GST pull-down assay confirmed direct interaction between GST-SDE15 and MBP-CsACD2 fusion proteins (FIG. **10**C). These results demonstrate a genuine interaction between SDE15 and CsACD2.

Example 9

C-Terminal is Essential for SDE15 to Interact With RCCR Domain of CsACD2

[0206] To further define the regions of SDE15 and CsACD2 involved in interaction, truncated proteins GST-SDE15\N (SDE15 protein without the signal peptide and additional 25 aa at the N-terminus), GST-SDE151C (SDE15 protein without the signal peptide and additional 25 aa at the C-terminus) and MBP-RCCR (the RCCR domain of CsACD2 with deletion of 71 aa at the N-terminus) were expressed in *E. coli,* purified and used in GST pull-down assays (FIG. **10**D, E). Results showed that the RCCR domain of CsACD2 was sufficient to interact with GST-SDE15 (FIG. **10**D). In addition, deletion of the C-terminal, but not the N-terminal of SDE15 abolished its interaction with CsACD2 (FIG. **10**E).

[0207] Next, we examined the subcellular localization of CsACD2 in *Nicotiana benthamiana* leaf cells. CsACD2 was mostly found in the chloroplast and partially in the cytosol and the nucleus (FIG. **11**). Co-localization of SDE15-CFP and CsACD2-EYFP could be detected in the cytoplasm and the nucleus (FIG. **11**).

Example 10

SDE15 Promotes the RCCR Activity of CsACD2

[0208] It has been shown that ACD2 in *Arabidopsis* represses PCD by functioning as a red chlorophyll catabolite reductase (RCCR) to catalyzing porphyrin-related molecules such as RCC Pruzinská, A., Tanner, G., Aubry, S., Anders, I., Moser, S., Müller, T., Ongania, K. H., Kräutler, B., Youn, J. Y., Liljegren, S. J., and Hörtensteiner, S. (2005). Plant Physiol 139, 52-63). To determine whether CsACD2 is involved in SDE15-mediated suppression of PCD, we infiltrated the leaves of *N. benthamiana* with *Agrobacterium* harboring the binary vector that expresses CsACD2-YFP fusion protein and/or SDE15, followed by infiltration with *Agrobacterium* containing a binary vector that expresses AvrBsT 2 days after CsACD2 infiltration (FIG. **12**A, B). As expected, AvrBsT triggered an HR in the absence of SDE15 but that HR was repressed in the presence of SDE15. On the other hand, we found neither SDE15.sup.ΔN nor SDE151.sup.ΔC could repress HR induced by AvrBsT. Intriguingly, however, we found that transient overexpression of CsACD2-YFP protein alone was also sufficient to suppress AvrBsT-elicited HR in *N. benthamiana* (FIG. **12**A, B), suggesting that (i) SDE15 could interact with the endogenous homolog of ACD2 in *N. benthamiana* (NbACD2 hereinafter) and (ii) it enhances the stability or activity of CsACD2. To test the first possibility, we performed GST pull-down assays. Indeed, SDE15 interacted strongly with NbACD2. As in the case of CsACD2, the NbACD2-YFP fusion protein alone, when transiently expressed, was sufficient to suppress HR induction by AvrBsT. To examine the second hypothesis, we directly performed RCCR activity assay. SDE15 significantly increased the RCCR activity of CsACD2 to catabolize RCC to primary fluorescent catabolite (pFCC), while SDE15 1.N and SDE151C lost the ability to promote the RCCR activity of CsACD2 (FIG. **12**C).

Example 11

SDE15 Promotes the Chlorophyll Break-Down in Planta

[0209] HLB is associated with tissue chlorosis and vein mottling, both are indicative of chlorophyll break down and previous studies have shown that the levels of chlorophyll a and b were significantly reduced in HLB-diseased *Citrus* trees (Killiny, N., and Nehela, Y. (2017). Mol Plant Microbe Interact 30, 543-556). As ACD2 is critical for alleviating the accumulation of toxic intermediates in the chlorophyll breakdown pathway, we surmise that SDE15 promotes the RCCR activity during Las infection by lowering toxic intermediates that induce PCD. We quantified the concentration of pheophorbide a, which is upstream of RCC in the chlorophyll break-down pathway. The concentration of pheophorbide a was significantly lower in the leaves of SDE15 transgenic *Citrus* than that of the EV transgenic *Citrus* under HLB-free conditions. In addition, the levels of chlorophyll a and chlorophyll b were also lower in SDE15 transgenic *Citrus* than those in the EV transgenic *Citrus*. Furthermore, the concentrations of the three compounds were lower in the leaves of Las-infected *Citrus* (both SDE15 and EV transgenic plants) than those in the healthy EV transgenic plants. Our results suggest that, by promoting the RCCR activity of CsACD2, SDE15 likely prevents accumulation of PCD-eliciting intermediates during the breakdown of chlorophylls. This activity also likely contributes to the development of yellowing symptom associated with HLB and explained the yellowing symptoms observed in the SDE15 transgenic *Citrus* (FIG. **6**B).

Example 12

Methods For Examples 5-11

Vectors Construction

[0210] To generate the construct for plant transformation, Las genomic DNA was isolated from HLB diseased *Citrus* leaf by CTAB method. The coding sequence of SDE15 (222 bp) without signal peptide was PCR-amplified using gene-specific primers (Table S2). A BamHI recognition sequence and a KpnI recognition site with two protecting nucleotides were added to the 5′ end of primers. The PCR product was purified and cloned into pGEM-T Easy vector (Promega, Madison, WI, USA) and then cloned into the binary vector erGFP-1380N at the BamHI and KpnI sites to generate SDE15-overexpression vector. The resulting binary vector was transferred into *Agrobacterium tumefaciens* strain EHA105 and LBA4404 for *Citrus* and tobacco transformation. Empty Vector (EV) without SDE15 fragment was used for *Citrus* and tobacco transformation as negative controls.

[0211] For Y2H, the coding region of SDE15 (minus the putative signal peptide) was amplified and cloned in-frame with the GAL4 DNA-binding domain (BD) of the bait vector pGBKT7 to generate BD-SDE15 for Y2H library screening and co-transformation in yeast. The 996 bp coding sequence of CsACD2 was PCR-amplified from *Citrus* leaf cDNA and cloned in-frame with the GAL4 DNA-activating domain (AD) of the prey vector pGADT7 to generate AD-CsACD2 for co-transformation with bait vector in yeast to confirm the interaction. BD and AD vectors were constructed by using the In-Fusion cloning kit (Clontech, Mountain View, CA, USA).

[0212] Transient expression in *Citrus* protoplast were used for subcellular localization and BiFC assays. For the subcellular localization assay, the coding region of SDE15 without signal peptide was inserted into EcoRI-digested C-terminal EYFP containing vector pSAT6-EYFP-NI by using In-Fusion cloning kit to generate SDE15-EYFP fusion proteins. For the BiFC assay, SDE15 and CsACD2 were inserted into SalI-digested BiFC vectors pSAT6-nEYFP-Cl and pSAT6-cEYFP-C1-B by using In-Fusion cloning kit to produce SDE15-EYFPN, EYFPC-SDE15, CSACD2-EYFPN and EYFPC-CsACD2 fusion proteins. All the vectors were subsequently used for *Citrus* protoplast transformation.

[0213] To generate recombinant protein constructs for GST pull-down assay and red chlorophyll catabolite reductase (RCCR) assay, the coding region of SDE15 (without signal peptide) was inserted between EcoRI and XhoI sites of pGEX-4T-1 vector (GE Healthcare, Chicago, IL, USA) to generate GST-SDE15 fusion protein vector as bait. The coding sequence of CsACD2 was inserted between BamHI and EcoRI sites of pMAL™-c5X vector (NEB, Ipswich, MA, USA) to generate MBP-CsACD2 fusion protein vector as prey and source of RCCR for enzyme assay. The truncated sequences of SDE15 fragments (SDE15.sup.ΔN and SDE15.sup.ΔC) were inserted into EcoRI-digested pGEX-4T-1 vector by using In-Fusion cloning kit to generate GST-SDE15.sup.ΔN and GST-SDE15.sup.ΔC fusion protein vectors as bait. The coding sequence of RCCR domain of CsACD2 was amplified and inserted between BamH I and EcoRI sites of pMAL™-c5X vector by using In-Fusion cloning kit to generate MBP-RCCR fusion protein vector as prey.

[0214] To generate the constructs for agro-infiltration assay in *N. benthamiana*, modified pCambia1380 vectors were constructed by inserting cauliflower mosaic virus promoter (CaMV 35S) and EYFP/CFP coding sequence to create the transient expression vectors with C-terminal EYFP reporter protein (pCambia1380-35S-EYFP) or C-terminal CFP reporter protein (pCambia1380-35S-CFP). The coding sequence of SDE15 without signal peptide was PCR-amplified and inserted between BamH I and Kpn I sites of pCambia1380-35S-EYFP and pCambia1380-35S-CFP to generate 35S-SDE15-EYFP and 35S-SDE15-CFP individually. The truncated sequences of SDE15 fragments (SDE15.sup.ΔN and SDE15.sup.ΔC) were inserted between BamH I and Kpn I sites of pCambia1380-35S-EYFP by using In-Fusion cloning kit to generate 35S-SDE15.sup.ΔN-EYFP and 35S-SDE15.sup.ΔC-EYFP. The coding sequence of CsACD2 was inserted between BamH I and Kpn I sites of pCambia1380-35S-EYFP to generate 35S-CsACD2-EYFP vector. The coding sequence of NbACD2 was inserted between BamH I and

Kpn I sites of pCambia1380-35S-EYFP by using In-Fusion cloning kit to generate 35S-BenACD2-EYFP vector. All the vectors were then transferred into *Agrobacterium tumefaciens* strain GV2260 for agro-infiltration assay.

[0215] All the primers used for vector construction were listed in Table S2.

TABLE-US-00003 TABLE S2 primers used for vector construction Forward (5′-3′) Reverse (5′-3′) SDE15-OE GGGGATCCATGGATACTCTCTCTGACTC GGGGTACCTCTTTCCCATTCTCTAAC BD-SDE15 CATGGAGGCCGAATTCATGGATACTCTCTCTGACTC GGATCCCCGGGAATTCTCTTTCCCATTCTCTAAC AD-CsACD2 GGAGGCCAGTGAATTCATGGCTGTGAACCACTTATG CACCCGGGTGGAATTCGGCAGTAAAAACCTTCTGTA SDE15-BiFC GAATTCTGCAGTCGACATGGATACTCTCTCTGACTC CCGCGGTACCGTCGACTCTTTCCCATTCTCTAAC CsACD2-BiFc GAATTCTGCAGTCGACATGGCTGTGAACCACTTATG CCGCGGTACCGTCGACGGCAGTAAAAACCTTCTGTA GST-SDE15 GGGAATTCATGGATACTCTCTCTGACTC GGCTCGAGTCTTTCCCATTCTCTAAC GST-SDE15.sup.ΔN TGGATCCCCGGAATTCATGGACGACTCCCATAATCAA GTCGACCCGGGAATTCTCTTTCCCATTCTCTAAC GST-SDE15.sup.ΔC TGGATCCCCGGAATTCATGGATACTCTCTCTGACTC GTCGACCCGGGAATTCTATATTGTTCTTTATCTTTAT MBP-CsACD2 TATCGTCGACGGATCCATGGCTGTGAACCACTTATG TACCTGCAGGGAATTCGGCAGTAAAAACCTTCTGTA MBP-RCCR TATCGTCGACGGATCCATGCCTGTTAGGCAGCTGAT TACCTGCAGGGAATTCGGCAGTAAAAACCTTCTGTA MBP-NbACD2 TATCGTCGACGGATCCATGGCTATTTCAATATCCT TACCTGCAGGGAATTCAGCATTGTAGATTTCCC SDE15-YFP TTGGATCCATGGATACTCTCTCTGACTC GGGGTACCTCTTTCCCATTCTCTAAC SDE15-CFP TTGGATCCATGGATACTCTCTCTGACTC GGGGTACCTCTTTCCCATTCTCTAAC SDE15.sup.ΔN-EYFP GGACTCTAGAGGATCCATGGACGACTCCCATAATCAA CCCTTGCTCACCATGGTACCTCTTTCCCATTCTCTAAC SDE15.sup.ΔC-EYFP GGACTCTAGAGGATCCATGGATACTCTCTCTGACTC CCCTTGCTCACCATGGTACCTATATTGTTCTTTATCTTTAT CsACD2-YFP TTGGATCCATGGCTGTGAACCACTTATG GGGGTACCGGCAGTAAAAACCTTCTGTA NbACD2-YFP GGACTCTAGAGGATCCATGGCTATTTCAATATCCT CCCTTGCTCACCATGGTACCAGCATTGTAGATTTCCC *Nucleotides underline is the restriction enzyme cutting site

Transient Gene Expression in Citrus Protoplasts

[0216] Protoplasts were isolated from etiolated Duncan grapefruit epicotyl segments by following the protocol of transient gene expression in *Arabidopsis* mesophyll protoplasts with modifications (Yoo, S. D., Cho, Y. H., and Sheen, J. (2007). Nat Protoc 2, 1565-1572). Briefly, epicotyl segments of Duncan grapefruit cultured in dark were cut to small pieces and digested in Cellulose “Onozuka” R-10 and MACEROZYME R-10 (Yakult Pharmaceutical, Japan) enzyme solution overnight. Protoplasts were harvested and used for plasmid transformation. Plasmids were transformed into *Citrus* protoplasts by the polyethylene glycol 4000 (PEG4000)-mediated transformation method (Citovsky, V., Lee, L. Y., Vyas, S., Glick, E., Chen, M. H., Vainstein, A., Gafni, Y., Gelvin, S. B., and Tzfira, T. (2006). J Mol Biol 362, 1120-1131; Lee, L. Y., Fang, M. J., Kuang, L. Y., and Gelvin, S. B. (2008). Plant Methods 4, 24). For the BiFC assay, the coding sequence of SDE15 without signal peptide and full-length CsACD2 were cloned into either N-terminal or C-terminal fragments of EYFP vectors pSAT6-nEYFP-C1 and pSAT6-cEYFP-C1-B (Citovsky et al., 2006). The combinations of SDE15-EYFP.sup.N/EYFP.sup.C-CsACD2 and CsACD2-

EYFP.sup.N/EYFP.sup.C-SDE15 were transiently co-transformed into protoplasts. Other combinations, such as SDE15-EYFP.sup.N/EYFP.sup.C, EYFP.sup.N/EYFP.sup.C-SDE15, SDE15-EYFP.sup.N/EYFP.sup.C-SDE15, CsACD2-EYFP.sup.N-/EYFP.sup.C, EYFP.sup.N/EYFP.sup.C-CsACD2, CSACD2-EYFP.sup.N/EYFP.sup.C-CsACD2, EYFP.sup.N/EYFP.sup.C were also transformed into *Citrus* protoplasts as controls. After incubation in dark overnight, the EYFP signals were examined and photographed under a fluorescence microscope for BiFC assay with excitation wavelength 514 nm (Olympus, Tokyo, Japan).

Plant Transformation and Pathogen Inoculation

[0217] *Agrobacterium* mediated transformation of etiolated epicotyl segments of Duncan grapefruit were carried out as described previously (Orbović, V., and Grosser, J. W. (2015). Methods Mol Biol 1224, 245-257). *Agrobacterium tumefaciens* EHA105 harboring the recombinant plasmid was used for *Citrus* transformation. Transgenic lines showing kanamycin-resistance and erGFP-specific fluorescence were selected and then micro-grafted in vitro onto 1-month old *Carrizo* citrange nucellar rootstock seedlings. After a month of growth in vitro, the grafted shoots were potted into a peat based commercial potting medium and acclimated under greenhouse conditions.

[0218] *N. tabacum* cv. Petite Havana SR1 seeds were sown on MS medium (Sigma-Aldrich, St. Louis, MO, USA) containing 3% sucrose and 0.8% agar and allowed to germinate at 22±1° C. (16h light and 8 h darkness). Subsequently, plants were grown and maintained in MS medium. Fresh tobacco leaf discs were infected with *A. tumefaciens* strain LBA4404 harboring the recombinant plasmid. The regenerated shoots were maintained on MS medium supplemented with 0.2 mg L.sup.−1 NAA and 3 mg L.sup.−1 6-BA along with 100 mg L.sup.−1 kanamycin and 500 mg L.sup.−1 cephotaxime. Kanamycin-resistant, erGFP and PCR positive shoots of T0 transgenic plants were selected, transferred to the greenhouse and maintained up to T2 generations, which were used for phenotype inspection and further analysis.

[0219] For the HLB pathogenicity assay, the SDE15-transgenic and EV-transgenic trees was inoculated with Las via grafting as previously reported (Li, J., Pang, Z., Trivedi, P., Zhou, X., Ying, X., Jia, H., and Wang, N. (2017). Mol Plant Microbe Interact 30, 620-630). Midrib DNA was isolated from the grafted trees monthly after grafting up to 4-month post grafting and used to quantify Las by Taqman qPCR with Primer/probe combination (CQULA04F-CQULAP10-CQULA04R) as described previously (Wang, Z., Yin, Y., Hu, H., Yuan, Q., Peng, G., and Xia, Y. (2006). Plant Pathology 55, 630-638). The Ct value of each amplicon represents the Las genomic copy numbers in 100 ng *Citrus* midrib DNA. The test was repeated three times.

[0220] For the *Xanthomonas citri* subsp. *citri* (Xcc) pathogenicity and hypersensitive reaction (HR) assays in *Citrus*, SDE15-transgenic and non-transgenic Duncan grapefruit plants were used for inoculation in a quarantine greenhouse. The wild-type strain Xac306 causes disease on grapefruit whereas the Xcc A.sup.w strain triggers hypersensitive reaction in grapefruit leaves7. Xcc strains were grown with shaking overnight at 28° C. in NB, centrifuged down, and suspended in sterile tap water, and the concentrations were adjusted to 10.sup.6 CFU/ml (for Xac306) and 10.sup.8 CFU/ml (for Xcc A.sup.w) individually. Bacterial solution was infiltrated into fully expanded, immature leaves with needleless syringes (Yan, Q., and Wang, N. (2012). Mol Plant Microbe Interact 25, 69-84.). The tests were repeated three times with similar results. Disease symptoms and HR phenotype were photographed at 3, 5, 7, 9 and 11 days post inoculation. Growth curve assay of Xac 306 was conducted at 0, 1, 3, 5, 7, 9 and 11 days post inoculation.

Agro-Infiltration Assay in *N. benthamiana*

[0221] *A. tumefaciens* strain GV2260 cells containing binary vectors were cultured overnight in LB medium with 50 μg ml.sup.−1 of rifampicin and 50 μg ml.sup.n−1 kanamycin and re-suspended in induction medium (10 mM MgCl2, 10 mM MES pH 5.6, 200 uM acetosyringone), and incubated at 25° C. with shaking for 4 h. The cultures were diluted to OD600 of 0.1 or 0.2. For each vector, three leaves of young *N. benthamiana* plants were infiltrated with diluted *A. tumefaciens* suspension as triplicates.

[0222] For the HR assay, young leaves of *N. benthamiana* were first infiltrated with *A. tumefaciens* cells containing binary vectors for SDE15-EYFP and/or CsACD2-EYFP by using a needleless syringe, kept in a greenhouse for 2 days and then infiltrated with another *A. tumefaciens* strain harboring the binary vector carrying the AvrBsT protein which can trigger HR as reported previously Kim, N. H., Choi, H. W., and Hwang, B. K. (2010). Mol Plant Microbe Interact 23, 1069-1082). Agro-infiltrated plants were kept in a greenhouse and HR were examined and photographed at 3 days post AvrBsT inoculation. For the electrolyte leakage assay, leaf discs of AvrBsT infiltrated plants at 2 days post infiltration were floated on deionized water with shaking. The conductivity of the solution was measured 4 h later using an Oakton™ Conductivity Benchtop Meters (Thermo Fisher, Waltham, MA, USA). The *A. tumefaciens* transformant cells harboring an empty vector were infiltrated into the leaves of *N. benthamiana* as controls.

[0223] For the localization and co-localization assay, CsACD2-EYFP was co-expressed with the plasma membrane localization-marker PM-CFP, the nucleus-marker CFP-nucleus or SDE15-CFP in leaves of *N. benthamiana*. *A. tumefaciens* strain GV2260 harboring the corresponding plasmids were infiltrated into leaves at OD600 of 0.2. Subcellular localization was inspected and photographed 1 day post infiltration.

Extraction of Phloem Sap Proteins

[0224] An optimized method of protein extraction from phloem sap was performed by combining two methods reported before (Hijaz, F., and Killiny, N. (2014). Collection and chemical composition of phloem sap from *Citrus sinensis* L. Osbeck (sweet orange). PLOS One 9, e101830; O'Leary, B. M., Rico, A., McCraw, S., Fones, H. N., and Preston, G. M. (2014). J Vis Exp.). Briefly, 10-20 cm (0.5 cm diameter) stems from Las infected and uninfected trees were collected. The bark area was stripped into two pieces and was manually removed from the twig. The inner part of the bark was rinsed with deionized water and dried with Kim wipes. Then the bark strips were cut into about 1-cm pieces using a sterile razor blade and placed in a 60-mL syringe filled with distilled water. Vacuum was applied for 5-15 seconds repeatedly to let water penetrated barks. Then the barks were dried with Kim wipes and placed in a 20-mL syringe, centrifuged in 50 mL falcon tube for 10 min at 4,000 g, at 4° C. The collected phloem sap was centrifuged at 15,000 g for 5 min. The supernatant was heating for 5 min at 95° C. in SDS gel-loading buffer for SDE15 detection with specific antibody.

RNA Isolation and Expression Analysis of HLB Associated Genes

[0225] qRT-PCR was performed to detect the expression of SDE15 in SDE15-transgenic plants (both in *Citrus* and tobacco) and in non-transgenic *Citrus* plants and psyllids. We also examined the expression of HLB marker genes in the SDE15-transgenic *Citrus* and PR genes in SDE15-transgenic plants after HR induction. Total RNA of transgenic *Citrus*, transgenic tobacco and psyllids were extracted by Trizol reagent (Thermo Fisher) and digested with DNase I (Promega) followed by the manufacturers' instructions. First-strand cDNA was synthesized from purified RNA with ImProm-II™ Reverse Transcription System (Promega) and diluted 10 times for RT-qPCR to detect related genes with specific primers (Table S3). 20 μl of qPCR reaction consisted of 10 μl of 2×KiCqStart® SYBR® Green qPCR ReadyMix™ (Sigma-Aldrich), 1 μl of each primer (5 μM), 2 μl of diluted cDNA template, and 6 μl of DNase/RNase free water. The PCR cycling consisted an initial activation step at 95° C. for 3 min, followed by 40 cycles of 95° C. for 15 s and 60° C. for 40 s. All cDNA samples were run in triplicates. Citrus GAPDH gene, tobacco Actin gene and Las gyrA gene were used as endogenous controls wherever appropriate.

[0226] The qPCR primer sequences of specific genes and endogenous control genes are listed in Table S3.

TABLE-US-00004 TABLE S3 primers used for qRT-PCR and Taqman probe PCR analysis Forward (5′-3′) Reverse (5′-3′) qSDE15 ACTCCCATAATCAAAAGCCTACG CGTATCTTTCACCATTCCATCCTC qCsACD2 GGCTAAATCAGTGTGCTTGTG

ATCAACCCATCCCTCTTTTCC PR1 AAATGTGGGTGAATGAGAAAGC ATTATTGTTGCACGTCACCTTG PR2 TTCCACTGCCATCGAAACTG GTAATCTTGTTTAAATGAGCCTCTTG PR3 GGCTCAAACTTCACATGAAACTAC GTTGACAATAATCTCCAGGGTTTC PR5 CACCATTGCCAATAACCCTAATG GGGACAGTTACCGTTAAGATCAG PP2-B15 TCGTTGCCATCAGAAGTATCAC CCAACGCAAATAAACTGTCCC WRKY40 CTCCTGTTCCAAATGCCAAG CCGAGGTGAGGGATTATCTTTAG ZIP5 TGAATATGCTGGTGAATCGGAG GCTGCAACCAAAGGCTTAATAG Sweet7 GCTAACCCTACTTCACTCCAC GGCATATACTCCACGCTCTTG Sweet15 GTGTTGCCGTTTCTGTTAGTG GCGAACCACATAATTGCACTC Chalcone synthase GCGTTCTAGTCGTATGCTCTG GCCAATGATTAAAGCTGCGG CsGAPDH GGAAGGTCAAGATCGGAATCAA GGAAGGTCAAGATCGGAATCAA NbActin CCTGAGGTCCTTTTCCAACCA GGATTCCGGCAGCTTCCATT gryA CAATGTGCTGGTCAATGGTG AATCTCCATCAAGGCATCCAG CQULA04 TGGAGGTGTAAAAGTTGCCAAA CCAACGAAAAGATCAGATATTCCTCTA (β operon primer) CQULAP10 FAM-ATCGTCTCGTCAAGATTGCTATCCGTGATACTAG-TAMRA (β operon probe, 5′-3′)

Yeast Two-Hybrid Library Screening and Interaction Analysis

[0227] Total RNA was extracted from leaves of Valencia sweet orange (HLB symptomatic (S, Las Ct value 25˜26 per 100 ng DNA), HLB asymptomatic (AS, Las Ct value 28˜30 per 100 ng DNA) and healthy (H, Las free)) by Trizol reagent (Thermo Fisher), digested by DNase I (Promega). mRNA samples were purified using the NucleoSpin® RNA kit (Clontech). ALL three types of mRNA samples were used to construct yeast two-hybrid libraries in the pGADT7-Rec vector using the Make Your Own “mate and plate” library system (Clontech) following the manufacturer's instructions and transformed in the yeast strain Y187 by using Yeastmaker™ Yeast Transformation System 2 (Clontech). The titer of each constructed library is more than 3×10.sup.8 which represents the good transformation efficiency. BD:SDE15 construct was transformed into Y2HGOLD yeast strain (Clontech) Library screening was performed according to the Matchmaker Gold yeast two-hybrid system protocol (Clontech). Standard positive controls (pGBKT7-53 and pGADT7-T; Clontech) and standard negative control (pGBKT7-Lam and pGADT7-T) were included. After mating between the Gold strain transformed with BD: SDE15 and the Y187 libraries, diploid yeasts were plated on synthetic dropout (SD)/-Leu/-Trp (DDO), SD/-Leu/-Trp/-Ade/-His (QDO) and SD/-Leu/-Trp/-Ade/-His plus X-α-gal and Aurcobasidin A (AbA) (QDO/A/X) agar plates to detect the activation of reporter genes HIS3, ADE2, MEL1 (for α-galactosidase activity) and AbA.sup.r (for Aurcobasidin A resistance).The fragments of positive diploid yeast were amplified by colony PCR with Matchmaker® Insert Check PCR Mix 2 (Clontech) and analyzed by electrophoresis on a 0.8% TAE Agarose/EtBr gel. The PCR products with single band were purified and sent for sequencing. The PCR products with multiple bands indicate the presence of more than one prey plasmid in a heterozygote cell. For this situation, plasmids were isolated from the heterozygote cells with multiple plasmids with Easy Yeast Plasmid Isolation Kit (Clontech) and transferred into *E. coli* for sequencing. BLAST was used to compare the inserts nucleotide sequences to the genome of sweet orange to identify corresponding proteins which interact with SDE15.

Recombinant Proteins Expression and GST Pull-Down Assay

[0228] *E. coli* cells expressing GST or GST fusion proteins were washed in PBS buffer and suspended with CelLytic B Cell Lysis Reagent (Sigma-Aldrich) to generate the cell lysates. After centrifugation, the cell lysates were incubated with glutathione agarose beads in accordance with the GST Protein Interaction Pull-Down Kit instructions (Thermo Scientific). The beads were washed to remove the unbound proteins and incubated with *E. coli* cell lysates expressing either MBP or MBP fusion protein for 1 to 2 h at 4° C. After washing four times, the beads were eluted with 10 mM glutathione, and the eluates were collected and immunoblotted using anti-MBP (NEB)

and anti-GST (Abcam, Cambridge, UK) antibodies.
Enzyme Assays
[0229] Coupled Pheophorbide a oxygenase (PaO)/RCCR assay was performed to test CsACD2 activity according to published procedures (Hortensteiner, S., Vicentini, F., and Matile, P. (1995). Chlorophyll breakdown in senescent cotyledons of rape, *Brassica napus* L.: Enzymatic cleavage of phacophorbide a in vitro. New Phytol. 129, 237-246; Wüthrich, K. L., Bovet, L., Hunziker, P. E., Donnison, I. S., and Hörtensteiner, S. (2000). Plant J 21, 189-198; Pruzinská et al., 2005). Thylakoids containing PaO were isolated and solubilized from senescent *Citrus* leaves as described previously (Hortensteiner et al., 1995). PaO was partially purified from solubilized membranes and used for enzyme assay (Rodoni, S., Vicentini, F., Schellenberg, M., Matile, P., and Hortensteiner, S. (1997). Plant Physiol 115, 677-682.). MBP-CsACD2 fused protein was expressed and purified with the pMAL protein fusion & purification system (NEB) as the source of RCCR. Briefly, assays (total volume of 50 μl) contained different combinations of PaO (equivalent to 0.5 g of tissue), *E. coli* (50 μg) protein extracts as a source of RCC-forming factor (RFF), and purified MBP-CsACD2 (1.5 μg) as the source of RCCR. The assays were supplemented with 0.5 mM pheide a, 10 μg ferredoxin (Fd), and a Fd-reducing system consisting of 2 mM Glc-6-P, 1 mM NADPH, 50 milliunits of Glc-6-P dehydrogenase, and 5 milliunits of Fd-NADP.sup.+ oxidoreductase. After 1 hour incubation at 25° C., reactions were terminated by the addition of 80 mL methanol. Formation of primary fluorescent chlorophyll catabolite (pFCC) was followed by reversed-phase HPLC with 36% (v/v) 50 mM potassium phosphate buffer, pH 7.0, in methanol as solvent. Activities are determined as integrated fluorescence units (320/450 nm) of pFCCs.
Quantification of Compounds Participating Chlorophyll Break-Down Pathway
[0230] Three compounds (chlorophylls a, b and pheophorbide a) in Chlorophyll break-down pathway were extracted and quantified as previously descripted (Garrido, J. L., Rodríguez, F., Campaña, E., and Zapata, M. (2003). J Chromatogr A 994, 85-92.), with modification. Briefly, leaf samples of SDE15 transgenic *Citrus*, EV transgenic *Citrus*, SDE15 transgenic *Citrus* infected with HLB and EV transgenic *Citrus* infected with HLB were collected and homogenized with 8 ml of 90% acetone and left for 16 hours at −10° C. All extracts were filtered through 25 mm, 0.2 μm GHP Acrodisc filters (Sigma-Aldrich) prior to injection. All sample preparations were done under subdued light. The standards of chlorophylls a, b and pheophorbide a were obtained from Sigma-Aldrich. All the standards and samples were followed by reversed-phase HPLC. Mobile phase consisted of (A) methanol, (B) 0.025M ammonium acetate and (C) acetone. A linear gradient from A-B (80:20, v/v) to A-C (80:20, v/v) was pumped during 15 min, followed by an isocratic hold at A-C (80:20, v/v) during a further 5 min. The flow-rate was 1 ml/min.

Claims

1. A plant comprising plant cells comprising a modification to an endogenous gene, wherein the polypeptide encoded by the endogenous gene interacts with a Sec-dependent effector (SDE) secreted by a bacterial species from the genus *Ca. Liberibacter,* wherein the modification knocks-down or reduces expression of the endogenous gene and/or interrupts interaction of the polypeptide with an SDE and confers resistance or tolerance to *Ca Liberibacter* infection in the plant relative to a plant of the same variety lacking the modification, and wherein the endogenous gene encodes PP2-B2/12 and the modification comprises a modification to the endogenous gene comprising; insertion of at least 1 nucleotide, a deletion of at least 1 nucleotide, and/or a substitution of at least 1 nucleotide.
2. The plant of claim 1, wherein the plant is *Citrus.*
3. The plant of claim 1, wherein the modification is made to at least one of SEQ ID NO: 30, 31, or a sequence comprising at least 95% identity therewith.
4. The plant of claim 1, wherein the plant is a solanaceous plant.

5. The plant of claim 4, wherein the plant is *Solanum tuberosum.*
6. The plant of claim 1, wherein the plant is a grapefruit tree, an orange tree, a sweet orange tree, a lime tree, citrumelo tree, trifoliate tree, reticulata tree, aurantiuma tree, lemon tree, a papeda tree, a pummelo tree or a mandarin orange tree.
7. The plant of claim 1, wherein the SDE is Las4025.
8. A seed that produces the plant of claim 1.
9. A plant part of the plant of claim 1, wherein the plant part comprises the modification.
10. A method of generating a modified *Citrus* plant comprising resistance or tolerance to infection by a bacterial species from the genus *Ca. Liberibacter;* the method comprising the steps of: (a) modifying a PP2-B2/12 gene of a *Citrus* plant cell such that expression of the PP2-B2/12gene is knocked-down or reduced and/or interaction of the polypeptide encoded by said PP2-B2/12 gene with a Sec-dependent effector (SDE) secreted by a bacteria species from the genus *Ca. Liberibacter* is reduced; and (b) regenerating the modified plant from said plant cell or a progenitor cell thereof, wherein said the plant comprises said modification.
11. The method of claim 10, wherein the plant is *Citrus*.
12. The method of claim 11, wherein the plant is a solanaceous plant.
13. A plant comprising resistance to *Ca. Liberibacter* infection produced by the method of claim 10.
14. The method of claim 10, effects of a Sec-dependent effector (SDE) secreted by a bacteria species from the genus *Ca. Liberibacter* are reduced.
15. The method of claim 14, wherein the SDE comprises Las4025.
16. The method of claim 10, wherein step (a) comprises a genome-editing technique.
17. The method of claim 16, wherein the genome-editing technique comprises a use of a nuclease, wherein the nuclease introduces a single-strand DNA break or a double-strand DNA break.
18. The method of claim 16, wherein the genome-editing technique comprises use of a Talen, a ZFN, meganuclease, or a CRISP/Cas system.